Subject-Bound Evidence Composition for Delegated AI Actions

Executable Indistinguishability Witnesses and Commit-Time Revocation

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Background: Delegated AI workflows may contain individually valid policy, evidence, runtime-state, authority, measurement, and effect artifacts while lacking an end-to-end basis for one action.

Objectives: We characterize what local verifier verdicts cannot express, test a non-compensable decision rule over five typed artifact predicates and a partial cross-layer subject schema, and evaluate commit-time revocation and typed first-failure transfer under stronger evidence classes.

Methods: Three executable laboratories cover signed policy-version/evidence replay, SAFE metric metamorphics, and a frozen 104-transaction composed corpus with runtime-state, delegation, and binding checks. Companion experiments add Ed25519 authority credentials, generated mutations, scheduled revocation, 104 transaction-bound TPM2 quotes under eight fresh software-TPM roots, 372 process-isolated durable revocation cases with disconnect/restart injection, an independent compiled native verifier, an exhaustive finite Java checker, a 135-case five-container/two-effect-instance fault replay, and typed transfer of a public first-party 21-case two-language artifact-verifier result through Python and SQLite paths. The verified sealed capsule plus a hash-recorded portability overlay is additionally executed twice on separate hosted job VMs reporting x86_64 and ARM64 for four declared lanes: policy replay, composed-corpus evaluation, typed transfer, and process-isolated durable revocation. The same four-lane contract and a non-destructive live TPM companion are executed on an identified Ubuntu x86_64 Hyper-V VM exposing a Microsoft vTPM.

Results: The main corpus contains eight denials in which all five artifact verifiers pass but the subject-binding condition fails. Point-only measurement, four-of-five majority, and artifact-validity ablations produce 2, 31, and 48 false allows on the designed corpus; stronger partial joins leave 3, 2, and 1 when they omit both time and profile, time only, and profile only. A local-verdict indistinguishability theorem generalizes the counterexample beyond majority voting. Native quote appraisal reproduces all 104 state classes, rejects 64/64 predeclared mutations, and preserves 104/104 composed decisions; the compiled path matches all 104 class/gate and 64 mutation outcomes. An atomic revocation-status/effect guard has zero safety errors and zero duplicate effects; all 96 disconnect/restart cases recover, while incomplete profiles expose false allows. In the five-container extension, 540 signed responses reverify, 4/4 fault cases recover through the second effect instance, and duplicate decision/effect keys remain zero. Typed transfer preserves 21/21 native-code rows; a separately implemented SQL path matches 125/125 rows and detects all 46 class mutations and omissions. Booleanization collapses 498 unordered within-corpus pairs of distinct rejection-code types. Both hosted repetitions verify 210/210 capsule members, satisfy 20/20 semantic assertions per architecture, and satisfy 11/11 cross-architecture comparison assertions. The identified zeus2 VM/OS satisfies 20/20 four-lane assertions, verifies 104/104 live transaction-bound vTPM quotes, rejects 64/64 predeclared quote mutations, and leaves no new transient or persistent TPM handle.

Conclusions: Local Boolean validity cannot substitute for typed composition over the implemented effect, resource, time, and profile coordinates and one commit boundary. The result is scoped to author-defined synthetic fixtures and the reported execution domains; it is not a deployed-system rate, physical-host attestation, hardware-root claim, human-agreement result, or interoperability claim. Optional external labelling and semantic panels would broaden construct validity, but they are not observations used by the scoped benchmark claim.

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1 Introduction

An agent may present a valid identity and execute a schema-valid tool call. Its machine may report an affirming attestation result. The action receipt may be correctly signed and internally well formed while reporting a claimed effect. A model-quality score may exceed a policy threshold. These statements do not entail that the specific effect was both authorized and supported by an applicable measurement.

The gap is compositional, and it is a classical one: component certificates do not compose into a system certificate without explicit discharge of the assumptions each certificate rests on (Rushby 2002). Each verifier here answers a scoped question:

- a policy verifier identifies which appraisal rule was active;
- an evidence verifier checks integrity and freshness;
- a state verifier consumes a typed appraisal result; the frozen source fixtures encode it structurally, and the native overlay re-derives it from transaction-bound TPM2 evidence;
- an authority verifier checks whether a delegation path covers an action, with a terminal effect check against the receipt inside its own case file;
- a measurement verifier checks profile applicability and inferential support.

Effect is not a sixth artifact with a sixth verifier. It is a subject that several artifacts describe at once, and no single verifier can see all the descriptions. The policy artifact carries the required effect and the required measurement profile; the evidence artifact carries the effect, resource, and issue time actually reported; the authority artifact carries the terminal narrowed scope and the interval in which the whole delegation path is valid; the measurement artifact carries the profile the appraisal was actually run under. A predicate over all four has no local home. The previous revision of this article asserted that a mismatch between the canonical action and the receipt's reported effect would surface as an authority or evidence failure. That statement was false, and the artifact proved it false: neither layer can see both sides of the comparison.

Local validity may coexist with global denial. For example, a signed but superseded policy can be authentic; a replayed receipt can retain a valid signature; a point estimate can exceed a threshold while its lower confidence bound does not; and an evidence receipt can validly report a WRITE effect when the valid delegation path permits only READ.

That last case is this article's worked example, and it is worked in both directions. In the corpus the previous revision shipped, the composed verifier **allowed** it: a correctly Ed25519-signed evidence object with effect `ledger.write` on resource `invoice-999` and an unseen nonce, evaluated under a policy whose required effect is `ledger.read` against an `ALLOW` delegation path whose terminal grant carries only `ledger.read` on `invoice-123`, returned policy `PASS`, evidence `PASS`, state `PASS`, authority `ALLOW`, measurement `PASS`, verdict `ALLOW`, and first rejecting gate `verified`. The composition thesis was asserted, not implemented. The present revision implements it with an explicit binding stage, and the same transaction now returns binding result `EFFECT_MISMATCH`, verdict `DENY`, and first rejecting gate `binding.effect` — with all five component verifiers still passing, and with a matched lookalike that differs only in the reported subject fields still allowed at gate `verified`. A standalone acceptance test builds both transactions from first principles rather than reading the frozen corpus, so the result cannot be satisfied by a corpus entry carrying a convenient label (Sections 4, 5.4, and 6.3).

This article evaluates the following decision rule:

$$\mathrm{Allow}(x) = P(x) \wedge E(x) \wedge S(x) \wedge A(x) \wedge M(x) \wedge B(x),$$

where P is exact policy applicability, E is evidence integrity/freshness, S is runtime state, A is authority-to-effect validity, M is measurement support, and B is the implemented partial binding. The rule is non-compensable: five passing predicates cannot offset one failed predicate. B is not a sixth artifact verifier and introduces no sixth artifact. It is a well-formedness condition on the conjunction over effect, resource, report time, and measurement profile fields extracted from the policy, evidence, authority, and measurement artifacts. The current B does not bind a state/workload subject or the other omitted coordinates listed in Section 4.1. The conjunction is commutative; the *evaluation* order is fixed separately and places B last (Section 4).

The work has three objectives:

1. establish the semantic limit of local validity and make its remedy executable, using explicit policy and measurement semantics, re-executed typed layer evaluators, a cross-layer binding stage, and matched indistinguishability witnesses;
2. test the two runtime boundaries abstracted by the main corpus: native appraisal of a frozen attestation vector and atomic revocation-status checking with effect commit; and
3. test whether the typed first-decisive-state representation transfers to a separately built artifact verifier while preserving its native diagnostic codes.

The result is an executable artifact study, not a claim of production readiness or population-level safety improvement.

2 Research Questions and Claim Boundaries

2.1 RQ1: When is local validity insufficient?

Can individually passing policy, evidence, state, authority, and measurement verifiers be locally indistinguishable between a transaction matched on the implemented coordinates and one mismatched on a required coordinate that must be denied? Within this question we test exact policy state, SAFE measurement semantics, typed composition, partial-binding ablations, named-gate localization, and the number of constructed all-local-pass denials in the frozen corpus.

2.2 RQ2: Where must runtime authority become atomic?

Does native offline appraisal preserve all 104 transaction state classes while rejecting targeted quote, PCR, freshness, and subject-binding mutations, and, in process-isolated signed-status races with durable retry and restart, does a linearizable status-and-effect guard avoid the false allows and duplicate effects exposed by incomplete protocols?

2.3 RQ3: What survives transfer across verifier ecosystems?

Can a separately constructed two-language artifact-verification experiment be represented in the same first-decisive-state form without discarding native rejection codes, and how much diagnostic distinction would a Boolean valid/invalid projection erase?

2.4 Claim ceiling

The study may report exact observations from frozen artifacts. It does not claim:

- prevalence in deployed agent systems;
- a defect in Veraison, A2A, MCP, RATS, SCITT, SAFE AI, or TOPSIS;
- ownership of the SAFE integration functions of Giudici and Kolesnikov;
- invention of authenticated or transitive delegation;
- externally independent labels;
- production interoperability or legal compliance;
- sufficiency of the proposed passport fields for every environment.

Three further non-claims are added in this revision, each traceable to the internal review of 2026-07-27:

- external independence of the expected labels: all labels are author-written within this research programme, and the composition-level labels are generated by the same shared composition-rule module the composed verifier imports, so agreement at the composition level is analytic (Section 6.3);
- cryptographic attestation in the **source corpus**: its original state predicate is a structural appraisal of a corpus-supplied attestation-result object, with no TPM, no Veraison, and no verification of any runtime. The r5 overlay is a separate replacement execution: it supplies transaction-bound TPM2 evidence for all 104 state decisions and recomposes the complete tuples, but remains software-TPM evidence on one host rather than attestation of a deployed workload;
- real-world false-allow rates: every false-allow count in this article is a count on a designed corpus, a corpus-coverage statistic, and must not be read as an estimate of prevalence anywhere.

A fourth is added for the binding stage introduced in this revision. The binding stage is a deterministic string-agreement and interval-containment condition over subject fields read out of corpus-supplied artifacts. It verifies no signature, takes no measurement, contacts no attestation service, and does not upgrade any corpus-supplied flag into a cryptographic fact. The denials it produces are denials on a designed corpus, and the claim it supports is representability and localization, not detection performance.

The retained r4 companions and the r5 replacement experiments introduce three further ceilings:

- the retained seven-case native companion is one frozen `swtpm` vector; the r5 replacement covers all 104 transactions under eight fresh software-TPM roots, but neither experiment supplies a hardware root, live attested workload, manufacturer endorsement, or independent-host evidence;
- the retained 276-case service is one-process loopback; the r5 replacement separates status, effect, and contender processes with durable linearization and recovery, and a five-container extension adds two interchangeable effect instances plus a deterministic fault proxy. Both remain on one physical host with centralized SQLite linearization rather than multi-host, Internet-scale, consensus, or performance evidence;
- cross-ecosystem transfer means that an explicit analytic crosswalk preserves native first-failure codes from two executable corpora; it does not mean semantic equivalence, protocol interoperability, or independent validation.

3 Related Work and the Remaining Composition Seam

3.1 Measurement integration

Kolesnikov's master's thesis proposes an integrated S.A.F.E.-AI compliance score built from accuracy, explainability, and robustness vectors, denoted RGA, RGE, and RGR, and studies arithmetic, geometric, root-mean-square, and TOPSIS integration (Kolesnikov 2025). Giudici and Kolesnikov develop this integrated SAFE measurement line in a journal article and working paper (Giudici and Kolesnikov 2025, 2026). Babaei and Giudici provide a statistical SAFE AI package (Babaei and Giudici 2025).

We do not propose a replacement metric. We test whether an evidence identity distinguishes aggregation, profile, reference-set, dependence, and confidence semantics that a scalar score alone omits.

3.2 Delegation and authorization

South et al. argue that AI agents require authenticated and auditable delegation, including agent-specific credentials and robust scope translation (South et al. 2025). WAVE already provides signed authorization graphs, transitive scoped delegation, revocation, and independently verifiable authorization proofs (Andersen et al. 2019). JEDI provides restricted key delegation across scope and time (Kumar et al. 2019). We note that these two systems share several authors from one research group, so they are related lines of work rather than two independent bounds on novelty; either alone suffices to bound any claim about delegation primitives.

The remaining experimental seam is the join from a valid authorization path to the canonical action and effect described by evidence. Our authority layer asks whether the terminal action remains entailed after delegation, freshness, status, scope, and binding checks — but it asks that question only of the objects inside its own case file. The join across artifacts belongs to the composition, and that is where this revision places it (Sections 4 and 5.4).

3.3 Policy versioning and change impact

The policy-state subquestion of RQ1 has prior art in policy analysis. Margrave verifies access-control policies and computes the semantic impact of policy changes (Fisler et al. 2005), and XACML 3.0 carries an explicit `Version` attribute on policies and policy sets (OASIS 2013). Those tools answer what a policy means and how an edit changes it. Our subquestion asks a narrower operational question: whether the decision procedure binds each decision to the exact policy identity, version, validity window, and digest that were active at decision time, so that an authentic but superseded or substituted policy cannot silently govern an appraisal. Versioning metadata existing in a policy language does not by itself make the verifier check it; the replay corpus tests the checking.

3.4 Agent benchmarks

TeamBench evaluates coordination under operating-system-enforced role separation (Kim et al. 2026). AgentRedBench includes underspecified-authorization attacks over SaaS integrations (Dingeto and Leeney 2026). ToolPrivacyBench evaluates field-to-tool authorization for private information (Hu et al. 2026). TraceSafe localizes mutated safety failures in tool trajectories (Y.-S. Chen et al. 2026). Heartbeat-Bound Hierarchical Credentials evaluates bounded hierarchical revocation (Deochake 2026). Li benchmarks authenticated account-level actions that leave the user's assigned task, in a high-performance-computing setting (Li 2026).

These works preclude broad claims such as the first unauthorized-action, role-isolation, trajectory-localization, or cascading-revocation benchmark. Our narrower object is typed composition: separately appraised artifacts that become invalid only when exact policy, evidence, state, delegation, measurement, and effect relations are joined.

3.5 Attestation and transparent evidence

RATS separates Evidence, Verifier appraisal policy, reference values, and Attestation Results (Birkholz et al. 2023). SCITT distinguishes signed statements, transparency registration, receipts, and later audit (Birkholz et al. 2026). A2A and MCP define agent interaction and tool-facing authorization contexts (Linux Foundation 2026; Model Context Protocol 2025). NIST's software and AI agent identity initiative identifies authorization, auditing, and non-repudiation as active infrastructure problems (NIST NCCoE 2026).

These mechanisms motivate typed boundaries. An attestation result is not an action grant; a transparency receipt is not proof of measurement adequacy; and an authenticated tool call is not necessarily within a delegated task. The composition question, what discharges the assumptions between such certificates, is the modular-certification problem in an agent setting (Rushby 2002).

3.6 Closest JAIR neighbours: artifacts, diagnosis, and norms

Belardinelli, Lomuscio, and Patrizi formalize artifact-centric multi-agent systems with data/lifecycle semantics and temporal-epistemic model checking (Belardinelli et al. 2014). Micalizio and Torasso diagnose expected-effect failures in distributed, partially observable multi-agent plans, including dependent failures (Micalizio and Torasso 2014). Criado treats norm monitoring as a resource-allocation problem under incomplete observations and proves properties of a resource-bounded monitor (Criado 2018). These are closer technical neighbours than credential syntax.

Our narrower object is a single authorization transaction at one decision boundary. We neither model-check an artifact lifecycle, diagnose a plan trajectory, nor choose which norms to observe. We show that a composer seeing only local Boolean verdicts cannot distinguish matched from mismatched transactions, then retain typed projections, the first decisive gate, and commit-time status. The distinction is summarized explicitly:

Work	Decision object	Partial observation	Cross-artifact identity	Commit-time status	Main guarantee/evaluation
Belardinelli et al. (Belardinelli et al. 2014)	artifact-centric MAS lifecycle	agent knowledge	relational artifact data	temporal lifecycle	finite abstractions and model checking
Micalizio and Torasso (Micalizio and Torasso 2014)	multi-agent plan execution	distributed action observations	dependencies between action failures	asynchronous trajectory	diagnosis of primary/secondary failures
Criado (Criado 2018)	monitored norm set	resource-bounded observations	norm/event applicability	monitor execution	monitor properties and simulations
this work	one delegated authorization transaction	local typed verifier outputs	explicit partial subject projections	atomic status/effect boundary	local-verdict theorem plus executable witnesses

Dell'Anna et al. revise conditional norms from execution traces (Dell'Anna et al. 2022), while Saisubramanian et al. mitigate side effects caused by incomplete open-world models (Saisubramanian et al. 2022). They remain relevant conceptual neighbours, but are not claimed as the closest technical precedents.

3.7 Typed verifier states and concurrent commit

The public first-party EATF Agent Evidence Package toolkit supplies a two-language artifact-verification example with an author-defined decision-path oracle and native first-failure codes in TypeScript and Python (Tyche Institute 2026). We use that result only to test representational transfer: its codes remain native and are crosswalked to a coarse shared ontology for analysis.

For revocation, the relevant systems property is linearizability: concurrent operations should appear in a single order consistent with their real-time precedence (Herlihy and Wing 1990). A status read followed by a separate effect is not an atomic operation, even when the read is fresh and signed. Our live-race laboratory therefore compares read-based profiles with a status-and-effect guard whose check and commit share one server-side linearization point. Native state appraisal uses the TPM 2.0 quote-verification interface but only over a frozen software-TPM vector (Trusted Computing Group 2026).

4 Decision-Transaction Model

The previous revision of this section advertised a record carrying fields named `canonical_action` and `observed_effect`. No such fields existed: a search of the shipped artifact returned zero occurrences of either name, and the composition consumed five scalar layer results and nothing else. Both are now real record fields, written by the corpus builder, re-derived by the verifier through the shared `composition_rule.subjects_of` extractor, and read by the binding stage. This checks serialization and plumbing, not independent subject extraction. What follows is the record that actually ships, field for field:

```
DecisionTransaction {
  id                          -- stable transaction identifier
  family                      -- one of the five corpus families
  policy_vector_id            -- selects the signed policy and its evidence
  evidence_variant_id         -- optional evidence override, else null
  measurement_fixture_id      -- selects the measurement fixture
  measurement_fixture_source  -- provenance of that fixture
  state_fixture_id            -- names the attestation-result fixture
  attestation_result {        -- the embedded appraisal input
    status, issued_at, expires_at,
    observed_digest, reference_digest, profile }
  authority_case_id           -- selects the typed delegation-path case
  canonical_action {          -- the action the artifacts jointly authorize
    effect, resource, operation,
    granted_tools, granted_resources,
    authorised_from, authorised_until,
    measurement_profile }
  observed_effect {           -- the effect as reported and appraised
    effect, resource, issued_at, measurement_profile }
  expected {                  -- the author-written expected label
    verdict, first_rejecting_gate, failed_layers,
    layer_results, binding_result, binding_gate }
  note                        -- optional prose, on the binding family only
}
```

Neither subject record is a new artifact, and neither introduces a sixth verifier. Every field is read out of an artifact that one of the five verifiers already inspects:

- `canonical_action.effect` and `canonical_action.measurement_profile` are the signed policy object's `required_effect` and `required_measurement_profile`;
- `canonical_action.resource` and `.operation` are the delegation case's terminal action;
- `canonical_action.granted_tools` and `.granted_resources` are the terminal narrowed scope, that is the grant carried by the last delegation edge, which the authority verifier has already certified to be a subset of every parent scope whenever it returns `ALLOW`;
- `canonical_action.authorised_from` and `.authorised_until` are the intersection of the per-edge validity windows, the interval in which the whole path is valid;
- `observed_effect.effect`, `.resource`, and `.issued_at` are fields of the signed evidence object whose signature the evidence layer verifies;
- `observed_effect.measurement_profile` is the profile identifier of the fixture the measurement layer actually appraised.

The corpus builder stores both records; the verifier re-derives them from the re-executed inputs and asserts field equality, reported as `subject_matches` in the shipped summary, so these are checked record fields rather than decoration. They remain corpus-supplied strings throughout, and comparing them is not attestation of anything.

Every layer returns a typed result rather than one undifferentiated Boolean, and so does the binding stage:

```
policy      in {PASS, MISSING, STALE, SUBSTITUTED, INVALID_SIGNATURE}
evidence    in {PASS, TAMPERED, REPLAY}
```

```
state       in {PASS, CONTRAINDICATED, STALE, REFERENCE_MISMATCH}
authority   in {ALLOW, DENY(gate, edge_id, edge_index)}
measurement in {PASS, FAIL_POINT, FAIL_LCB, PROFILE_MISMATCH}
binding     in {PASS, EFFECT_MISMATCH, RESOURCE_MISMATCH,
                TIME_MISMATCH, PROFILE_MISMATCH, UNDETERMINED}
```

The evaluation order is fixed, and the binding stage is last:

```
policy -> evidence -> state -> authority -> measurement -> binding
```

All five layers are appraised to preserve a typed failure set. The `first_rejecting_gate` follows the fixed order and supports deterministic localization. It is not a claim that later failures did not exist.

Two properties of the binding stage belong here because they are properties of the record; the rule itself is given in Section 5.4. First, `failed_layers` deliberately remains a list over the five artifact layers only. The binding stage never appears in it and has no row or column in the co-failure matrix, so a transaction whose `failed_layers` is empty and whose `verdict` is `DENY` is exactly a cross-layer denial. Second, the binding function is pure over the two subject records and is evaluated for every transaction regardless of the layer results, while the gate ladder consults its outcome only after all five layers have passed. This is not fail-open: a transaction with a failed layer is already a `DENY`, and the earlier gate is the honest localization. When a required subject field is absent — for example when a `MISSING` policy supplies no `required_effect` — the binding result is `UNDETERMINED`, which never counts as a pass.

4.1 The formal spine: local-verdict indistinguishability

Let a_P, a_E, a_S, a_A, a_M be the five artifacts and let $V_i(a_i) \in \{0, 1\}$ be their local acceptance predicates. Each accepted artifact also exposes a partial subject projection $\pi_i(a_i)$: effect, resource, interval, measurement profile, or some subset. Define

$$L(x) = \bigwedge_{i \in \{P,E,S,A,M\}} V_i(a_i)$$

and let $B(x)$ be the conjunction of the cross-artifact equality, membership, and interval constraints in Section 5.4. The strict decision is $C(x) = L(x) \wedge B(x)$. A Boolean-only composer observes $v(x) = (V_P, V_E, V_S, V_A, V_M)$ but not the projections.

The implemented B is deliberately a **partial protected-subject schema**: it binds the policy-required effect and measurement profile to the signed reported effect, the authority action/scope resource, and the report `issued_at` to the authority interval. It does not yet bind principal, request identifier, full action parameters/digest, state workload identity, measured model/dataset identity, actual external effect time, or commit receipt. The theorem applies when a required global relation is not observable through $v(x)$ and a matched/mismatched pair exists; it does not automatically establish every omitted field. The executable 104-case result establishes only the four implemented relations. We use “subject-bound” below only in that bounded sense.

Theorem 1 (local-verdict indistinguishability). Suppose there are a matched transaction x^+ and a mismatched transaction x^- such that

$$v(x^+) = v(x^-) = (1, 1, 1, 1, 1), \qquad B(x^+) = 1, \quad B(x^-) = 0.$$

Every deterministic composer f whose only input is $v(x)$ returns the same decision for x^+ and x^-. Consequently, it cannot both allow the matched transaction and deny the mismatched transaction.

Proof. Equality of the observed vectors gives $f(v(x^+)) = f(v(x^-))$. The required decisions differ, so either the matched transaction is falsely denied or the mismatched transaction is falsely allowed. □

This is an information limitation, not a defect specific to majority voting. The corpus provides eight x^- witnesses and six matched x^+ controls, including a locally valid signed report for `ledger.write` paired with a locally valid terminal grant for `ledger.read`. Their existence is constructed evidence under the partial subject schema defined below, not a deployment-frequency estimate.

Corollary 1 (strict refinement). $C(x) \Rightarrow L(x)$, and the refinement is proper whenever a witness x^- from Theorem 1 exists. This follows directly from $C = L \wedge B$.

Proposition 2 (failure of compensating composition). For any critical predicate q omitted by a compensating rule, a transaction with $q(x) = 0$ and all counted predicates equal to one is accepted by that rule and denied by C. On this corpus the four-of-five false-allow count is an algebraic identity:

$$N_{\text{FA,majority}} = N_{\text{single local failure}} + N_{\text{all local pass, binding fail}} = 23 + 8 = 31.$$

The 2 and 48 counts of the other weak ablations, and the 3/2/1 counts of the partial subject joins, depend on which fixtures the corpus contains; they are implementation observations on the frozen corpus, not algebraic properties and not rate estimates.

4.2 Decision clock and the revocation linearization point

A decision transaction has more than one relevant time. We make that explicit with a decision clock

$$\kappa(x) = (t_P, t_E, t_S, t_A, t_M, t_B, t_C),$$

where the first six components are the policy, evidence, state, authority, measurement, and binding appraisal instants and t_C is the effect-commit instant. Each artifact contributes its own validity interval or freshness constraint. A revocation event has an effective instant t_R. Wall clocks, however, are insufficient to order concurrent requests reliably, so the live laboratory records a server-side linearization index ℓ for every status read, revocation, guarded commit, and unguarded effect.

Proposition 3 (atomic revocation safety under the laboratory model). If a revocation operation and a guarded effect commit are linearizable operations on the same credential state, and the guarded commit writes the effect only when that state is active at its linearization point, then every committed effect was active at commit. A separate status read cannot provide this guarantee: the admissible order

$$\ell_{\text{read}} < \ell_{\text{revoke}} < \ell_{\text{effect}}$$

returns an authentic active status and still commits after revocation.

Argument. The guarded operation holds the state lock across check and commit, so either it linearizes before revocation and commits while active, or revocation linearizes first and it denies. The three-operation schedule above is a counterexample for every profile whose check and effect have different linearization points. This is a property of the experimental service semantics, not a universal deployment theorem.

A dependency-free Java model checker separately exhausts the bounded models used by Theorem 1, Corollary 1, and Proposition 3. It imports no Python adapter, corpus builder, SQL oracle, expected-label module, or saved verdict stream. It enumerates both possible outputs of a Boolean-only composer on the shared all-pass vector; all 16 agreement states over effect, resource, time, and measurement profile; all six orders of one read, revoke, and commit; and all four topologically valid two-read schedules. This is implementation-diverse finite enumeration, not a proof of arbitrary programs or distributed systems.

4.3 What a Boolean verdict destroys

Let a typed verifier expose K distinct rejection states. The scalar map $s(r) = \text{REJECT}$ for every rejecting r is non-injective and erases all $K(K-1)/2$ pairwise distinctions among them. This count is independent of case frequency;

entropy and best-guess accuracy additionally depend on the fixture mix. In the transfer laboratory EATF has 16 native rejection states and the composed corpus has 28, so Boolean scalarization erases $16(15)/2+28(27)/2 = 498$ within-corpus pairwise distinctions. The analytic crosswalk retains every native code and adds a coarse comparison class; it does not replace the code.

5 Methods

The study design is summarized before the laboratory details so that each question has one primary artifact, observation, and claim ceiling.

RQ	Primary artifacts	Oracle or invariant	Primary observation	Ceiling
RQ1: local-validity limit	16 policy/evidence vectors; SAFE metamorphics; 104 composed transactions including 14 binding cases	typed precedence, metamorphic relations, re-executed layer evaluators, matched local-verdict witnesses	policy and measurement gate sensitivity; eight all-local-pass denials; false allows of weak and partial-binding ablations	constructed data; shared-rule composition labels are analytic
RQ2: runtime atomicity	104 TPM-bound state vectors + 64 mutations; 372 durable service cases	native TPM verification and frozen state oracle; transactional linearization and idempotency	104/104 state classes, 64/64 mutation rejects, 104/104 recompositions; safety, recovery, and duplicate-effect invariants	eight software-TPM roots and separate services, but one x86_64 physical host
RQ3: typed transfer	21 EATF rows + 104 transaction rows	native codes preserved by total crosswalk; separate SQL execution and mapping mutation	code coverage, crosswalk sensitivity, and scalarization loss	author-defined analytic mapping; no external semantic validation or interoperability

5.1 Policy-Version x Evidence Replay

This laboratory was named “Policy–State Replay” in the previous draft. It is renamed here because “state” in that name denoted the policy and evidence condition (missing, stale, substituted), while the predicate S in the decision rule denotes runtime state. In this article the word “state” is reserved for the runtime predicate.

We generated a deterministic 16-vector corpus: the full factorial

$$\{\text{correct, missing, stale, substituted}\} \times \{\text{good, tampered, replayed}\},$$

plus four gate-isolation vectors added in this revision, each carrying good evidence and failing exactly one strict policy gate:

- **version-only:** correct policy, version 1, active interval;
- **freshness-only:** correct policy, version 2, expired interval;
- **digest-only:** correct identity, version, and interval, one semantic field changed and re-signed, so only the digest comparison rejects;

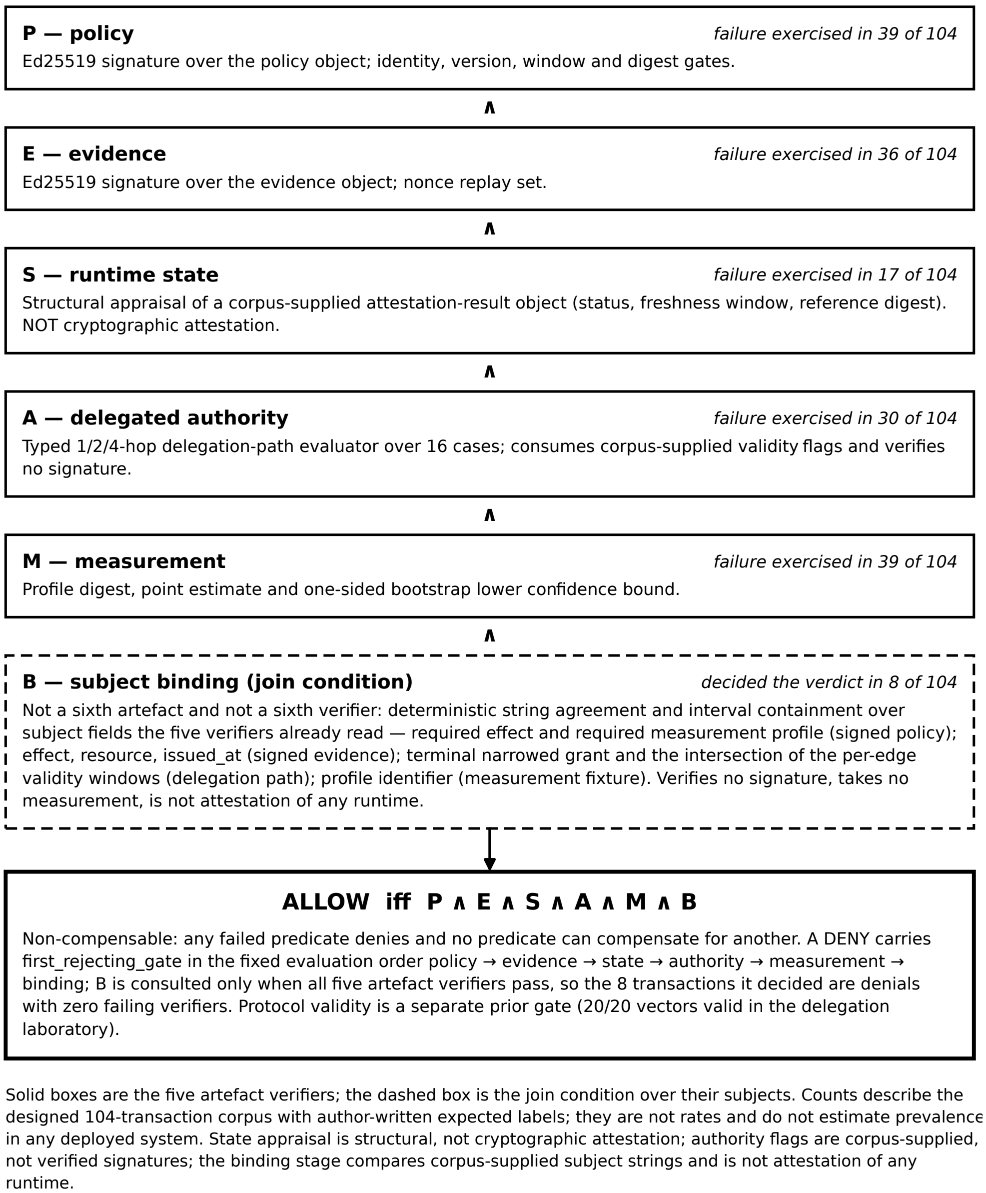


Fig. 1. The composition rule. Each of the five predicate nodes is annotated with the mechanism that establishes it in this artifact packet and with how often that layer fails across the 104 frozen transactions; the sixth node is the binding stage, which is not a layer and is annotated instead with the number of transactions whose verdict it decided. The counts describe corpus composition, not any deployed system; the state mechanism is a structural appraisal of corpus-supplied objects, not cryptographic attestation, and the binding stage compares corpus-supplied subject strings and verifies no signature.

- **invalid policy signature:** the policy object tampered after signing.

The additions close three coverage defects of the previous corpus, in which the version and freshness gates were fused in a single stale vector, the digest branch was unreachable, and a policy `INVALID_SIGNATURE` never occurred. Each strict gate (signature, identity, version, time, digest) is now exercised in isolation by at least one vector, so the sentence "the strict verifier checks policy signature, identity, version, time, and digest" is true of the experiment and not only of the source code.

Policies and evidence are signed with a deterministic Ed25519 test key. The key is public corpus material. The factorial vectors instantiate:

- **correct policy:** expected identity, version 2, active interval, and exact unsigned-object digest;
- **missing policy:** no policy object;
- **stale policy:** valid signature, version 1, and an expired interval;
- **substituted policy:** valid signature but a different policy identity and required effect;
- **good evidence:** valid signed effect with an unseen nonce;
- **tampered evidence:** effect changed after signing;
- **replayed evidence:** valid signature with a pre-seen nonce.

We compare the strict verifier with two experimental ablations:

1. signed-policy presence only;
2. missing-policy fail-open plus signed-policy presence.

A second, relational re-composition code path recomposes the verified policy and evidence layer states in SQL. It consumes the per-layer typed results emitted by the Python evaluator and re-derives only the conjunction and the first-rejecting-gate label; it does not reimplement Ed25519 and is not independent validation.

The experiment is motivated by a previously recorded Veraison freshness run in which Veraison correctly rejected a replay while the intended policy was active; the missing-policy and stale-policy fail-open cases occurred in this programme's own provisioning harness, not in Veraison. That record is an internal companion artifact; a public deposit is pending.

5.2 SAFE Metric Metamorphics

This laboratory is frozen from the previous packet and is consumed read-only in this revision; none of its corpus, code, or results changed.

For positive vectors a, b, and c, we evaluated tensor means of:

$$M_A(x, y, z) = \frac{x + y + z}{3},$$

$$M_G(x, y, z) = (xyz)^{1/3},$$

$$M_R(x, y, z) = \sqrt{\frac{x^2 + y^2 + z^2}{3}}.$$

The arithmetic and geometric tensor means have the factorized identities:

$$V_A = \frac{\bar{a} + \bar{b} + \bar{c}}{3},$$

$$V_G = \overline{a^{1/3}}\,\overline{b^{1/3}}\,\overline{c^{1/3}}.$$

policy \ evidence	**good**	**tampered**	**replayed**
correct	**ALLOW** **verified**	DENY evidence.integrity	DENY evidence.replay
missing	☐ DENY policy.missing	DENY policy.missing	DENY policy.missing
stale	☐ DENY policy.stale	DENY policy.stale	DENY policy.stale
substituted	☐ DENY policy.substituted	DENY policy.substituted	DENY policy.substituted

gate-isolation vectors (each fails exactly one strict policy gate):

☐ `version_only (good evidence)`	DENY policy.stale
☐ `freshness_only (good evidence)`	DENY policy.stale
☐ `digest_only (good evidence)`	DENY policy.substituted
`policy_sig_invalid (good evidence)`	DENY policy.signature

Strict typed verdicts. 16/16 agreement with author-written expected labels; the relational re-composition agrees 16/16 on verdict and rejecting gate, which is a second composition code path, not independent validation. This is reproduction of a frozen specification on designed vectors, not detection performance on any external workload.
☐ marks a strict deny that a weaker ablation profile falsely allows: signed-policy presence 5 of 15 denies, missing-policy fail-open 6. Ablations are author-defined experimental profiles, not named external products.

Fig. 2. Typed strict verdicts for the 16-vector Policy-Version x Evidence Replay corpus: the 4 x 3 factorial and the four gate-isolation vectors. Expected labels are author-written; 16/16 agreement demonstrates reproduction of a frozen specification on designed vectors, not detection performance on any external workload.

The suite uses a pure-Python loop implementation and a NumPy tensor transliteration of the same specification. Agreement between them excludes vectorisation and floating-point error, not specification error. The method is metamorphic testing (T. Y. Chen et al. 1998; Segura et al. 2016): each check states a relation that must hold between outputs rather than an oracle for one output. The fifteen checks divide into nine implementation identities (two-implementation agreement for the three integration operators and for TOPSIS, the arithmetic and geometric factorizations, the ordering $V_G \leq V_A \leq V_R$, and axis and within-axis permutation invariance) and six informative checks (tensor-versus-aligned-diagonal non-equivalence, profile-digest sensitivity under six semantic mutations, dynamic-reference TOPSIS rank behaviour, fixed-reference TOPSIS rank behaviour, measurement-fixture typing over four fixtures, and the sampling-overlap boundary fixture).

The synthetic confidence fixtures use shared three-column samples. For each fixture we compute a point estimate and a one-sided bootstrap lower bound $LCB_{1-\alpha}(C)$. The strict measurement rule requires:

$$\text{profile digest matches} \wedge \widehat{C} \geq \tau \wedge LCB_{1-\alpha}(C) \geq \tau.$$

The bootstrap is fully disclosed here because the boundary decision depends on it: the lower bound is the plain percentile method, 5,000 deterministic replicates, NumPy PCG64 generators seeded per dataset at 101, 102, and 103, with the resampling unit being the row of the shared sample. The fixture sample sizes are n = 30 (supported), n = 18 (point-only boundary), n = 30 (unsupported), and n = 30 (altered profile, which reuses the supported dataset). No bias-corrected or accelerated (BCa) interval was computed and no seed or replicate-count sensitivity sweep was run; both are limitations, and the percentile method's small-sample bias at n = 18 is exactly where the boundary fixture sits. Bootstrap validity is also dependence-sensitive (Cameron et al. 2008; Künsch 1989; Politis and Romano 1994), which motivates the overlap fixture below. The fixtures were designed to test boundary behavior and are not measurements of a deployed model.

The TOPSIS fixture contains alternatives A, B, and C plus one added alternative. Dynamic observed-reference TOPSIS recomputes normalization, ideal, and anti-ideal points from the current set. The fixed-reference variant uses a declared normalized 0–1 reference.

5.3 Authority-to-effect evaluator

The authority layer reuses the typed delegation-path suite from the protocol-valid-but-unauthorized laboratory of this packet. It constructs 1/2/4-hop paths and evaluates:

- lineage and role transitions;
- issuer-signature flags;
- permitted subdelegation;
- monotone scope;
- status and freshness;
- terminal action coverage;
- effect time and action binding.

The suite was extended in this revision from 12 to 16 typed cases. The four new cases each isolate one gate (`protocol.valid`, `edge.signature`, `effect.native_validity`, `effect.time_binding`) that was previously a dead branch in both the Python evaluator and the relational oracle.

What those four cases establish must be stated exactly, because the previous revision overstated it. Three of them — `D1_protocol_invalid`, `D2_signature_invalid_second_edge`, and `D1_receipt_native_invalid` — work by setting to `false` the very corpus-supplied Boolean that the gate under test reads (`protocol_valid`, `edges.1.issuer_signature_valid`, `receipt.native_evidence_valid`). Flipping the input of a branch and observing that both implementations take the branch and localize it to the same edge identifier and index is *branch coverage*, and coverage of two implementations at once. It is not verification of the property the branch names: no signature is checked, no protocol run is validated, and nothing about the flag's truth is established. Saying that these cases "close the blind spot" would confuse a live branch with a verified property. What they close is the possibility that the branch is unreachable and therefore silently untested in both implementations at once — which is a real and previously open defect, and a smaller claim. Only the fourth case, `D1_receipt_effect_time_outside_window`, exercises a computed comparison: it moves the receipt's effect time outside the delegation window, and the evaluator performs the interval test rather than reading a Boolean.

The suite reports 16/16 expected typed tuples, 13/13 first-invalid-stage matches on the negative cases, and 8/8 delegation-edge ID and index matches under an edge-gate definition that counts only cases rejected by an `edge`.

`*` gate. The same laboratory also retains a 20-vector semantic corpus (20/20) and a seven-case hybrid adapter over recorded A2A evidence (7/7).

The cryptographic component of this layer is stipulated, not computed: the fields `issuer_signature_valid`, `native_evidence_valid`, and `protocol_valid` are corpus-supplied experimental flags, and no cryptography executes anywhere in the authority laboratory. The shipped summary fields were renamed in this revision (for example `native_object_flags_true`) so the artifact says the same thing the prose does, and the disclaimer is embedded in the result JSON, not only here.

The delegation-path suite agrees 16/16 with an independent relational SQL implementation. We keep the word "independent" for this oracle alone, and it differs in kind from the composition-level SQL of Sections 5.1 and 5.4: it is a 233-line relational re-derivation that starts from the raw case objects, normalises edges, scopes, and receipts into tables, computes a violation set, and ranks violations with a window function, a set-then-rank formulation confronting the Python evaluator's imperative short-circuit. The case expansion and expected labels remain shared with the Python entry point, so even this agreement is implementation diversity over verdict derivation, not independent validation of the labels.

5.4 Composed frozen corpus

The composed corpus contains 104 transactions:

Family	Transactions	Purpose
policy/evidence/measurement factorial	36	cross the 12 signed replay vectors with three measurement outcomes
state–authority–measurement cube	8	enumerate all binary pass/fail combinations of those three layers
authority path reuse	16	re-execute all 16 typed 1/2/4-hop cases with other layers valid
cross-layer joins	30	10 layer pairs x 2 two-fault joins, 4 triples, 1 quadruple, 1 all-five failure, and 4 matched all-valid lookalike positives
cross-layer binding	14	8 transactions in which all five layers pass and the subject records disagree, each paired with a matched all-valid lookalike (6 distinct controls)

Two of these families are new relative to the 56-transaction predecessor, and they do two different jobs that the previous revision conflated under the word "join".

The cross-layer-joins family populates the co-failure structure. The predecessor corpus was block-diagonal, with zero co-failures between authority and policy, authority and evidence, state and policy, and state and evidence, so no transaction exercised two layers failing at once across those pairs. The joins family fills every off-diagonal cell of the 5 x 5 co-failure matrix, with a corpus-wide minimum off-diagonal count of 5, and its four matched all-valid lookalike positives ensure that universal refusal cannot masquerade as correctness. A populated co-failure cell is a co-occurrence of two independently caused layer failures in one transaction. It is not a join in the relational sense, and it does not test whether the layers are describing the same action.

The cross-layer-binding family, new in this revision, tests agreement over the implemented effect, resource, report-time, and measurement-profile coordinates; it does not test the omitted state/workload, principal, model, or commit-receipt coordinates. Each of its 8 denials is a transaction in which every one of the five component verifiers returns a pass and the composition denies because one implemented coordinate disagrees; each is paired with a matched all-valid lookalike that is allowed, so the denials are not universal refusal. Its evidence objects are owned by this laboratory and signed with the policy laboratory's own test key and signing function, imported rather than transcribed; every one carries a fresh nonce and a valid signature, so the unmodified evidence evaluator returns `PASS` on all of them and they differ only in the subject fields the binding stage reads. Its two extra measurement fixtures are likewise owned by this laboratory, in the SAFE fixture schema, appraised by the unmodified SAFE evaluator, which passes both; they differ from each other only in profile identity, and they exist because the signed policy cannot be varied without changing its digest, which would make the policy layer report `SUBSTITUTED`. Each transaction's `measurement_fixture_source` records which population its fixture came from. All these constraints are enforced as deterministic assertions at corpus build time and re-checked by the verifier's exit gate.

The binding stage itself is a single pure function over the two subject records, applied in a fixed rule order, first failing rule winning:

```
0. any required subject field absent            -> UNDETERMINED
1. signed reported effect differs from the policy-required effect,
   or is not in the terminal granted tools      -> EFFECT_MISMATCH
2. reported resource differs from the canonical action resource,
   or is not in the terminal granted resources  -> RESOURCE_MISMATCH
3. issued_at outside the authorised window      -> TIME_MISMATCH
4. measurement profile differs from the
   policy-required profile                      -> PROFILE_MISMATCH
5. otherwise                                    -> PASS
```

Each non-`PASS` outcome maps to the correspondingly named gate in the `binding.*` namespace. Timestamps are compared lexicographically, which is sound only because every timestamp in this artifact is fixed-width UTC `Z` form; both window boundaries are inclusive, matching the state adapter and the policy validity window. The time test is containment and never equality, and the reason is a real limitation that this revision reports rather than repairs: the three decision-relevant time constants in the artifact — the policy and state decision time 2026-07-25T21:30:00Z, the authority effect time 2026-07-25T20:00:00Z, and the evidence issue time 2026-07-25T21:29:30Z — are unrelated. The main corpus therefore cannot instantiate the decision clock of Section 4.2 from those constants. The live companion does instantiate its critical order with server-side linearization indices.

The stage is placed in the composition rather than inside a layer because a predicate over four artifacts has no local home. Putting it inside the evidence evaluator would make the evidence column of the co-failure matrix partly a copy of the policy column, and could not express the resource, time, or profile mismatches at all; putting it inside the authority evaluator would be cross-layer analysis wearing a single-layer costume and would make the composition rule vacuous. Only the join can see all four subjects, so only the join can reject on their disagreement.

A standalone acceptance test rebuilds the Section 1 counterexample and its lookalike from first principles and does not read the frozen corpus, so the result cannot be satisfied by a corpus entry carrying a convenient label. It asserts, and prints, that the shipped evidence evaluator still returns `PASS` on the WRITE receipt — the root cause is unchanged, and the fix is at the join and not in the layer — that all five artifact verifiers pass, that the binding result is `EFFECT_MISMATCH`, that the verdict is `DENY` at gate `binding.effect`, and that the lookalike is `ALLOW` at gate `verified`.

Two provenance properties are central and are declared inside the corpus and every generated summary. First, in the previous draft the five-layer precedence ladder existed as three same-author transcriptions (corpus builder, verifier, SQL oracle), so composition-level agreement could fail only if one transcription diverged from the others. This revision refactors the conjunction, the binding stage, and the first-rejecting-gate ladder into one shared composition-rule module imported by both the corpus builder and the verifier; the duplicated ladders were deleted. The consequence is stated where the numbers are reported: composition-level expected labels are generated by the same composition rule the verifier imports, so agreement at that level is analytic. Per-layer agreement is the non-analytic execution check; subject-record agreement is a same-function round trip. Second, transactions embed `attestation_result` objects, and the verifier calls a state adapter at the fixed decision time 2026-07-25T21:30:00Z (asserted equal to the policy corpus decision time) rather than comparing a corpus string. The adapter is a structural appraisal of the supplied object: a status gate, an inclusive freshness window, and a reference-digest comparison, in a fixed order, emitting `PASS`, `CONTRAINDICATED`, `STALE`, or `REFERENCE_MISMATCH`, and failing closed on missing fields. It performs no cryptographic verification, involves no TPM and no Veraison, and is not remote attestation; the disclaimer travels inside the result records. A 24-case adapter self-test runs as the first gate of the laboratory's entry point.

Corpus construction records the SHA-256 of each source corpus. The composed verifier verifies those hashes and re-executes the three source evaluators per transaction through `importlib`; it does not copy stored verdicts. The SAFE source corpus is consumed read-only from the frozen previous packet.

5.5 Ablations

We compare the strict composition with:

- **artifact validity / policy presence:** signed-policy presence and evidence integrity, ignoring policy identity, state, authority, measurement, and binding;
- **point-only measurement:** strict policy, evidence, state, authority, and binding, but $\widehat{C} \geq \tau$ without confidence or profile enforcement;
- **four-of-five majority:** allow when any four of the five artifact layers pass;
- **effect/resource binding:** require all five local verifiers plus effect and resource agreement, but omit time and measurement-profile agreement;
- **effect/resource + time:** add authorized-window containment, but omit measurement-profile agreement; and
- **effect/resource + profile:** add measurement-profile agreement, but omit authorized-window containment.

The two compensating ablations differ in what they remove, and this revision keeps that difference clean. The point-only profile replaces exactly one thing, the measurement criterion, and retains the binding stage; that is what makes it an ablation of one thing, and it is why the binding function is evaluated for every transaction rather than only when the layers pass. The four-of-five majority is by construction also an ablation *of the binding stage*: it votes over the five artifact verifiers, and a stage that is not one of them cannot appear in the vote.

These are intentionally incomplete experimental profiles, not representations of named products. The latter three are credible subject-aware schema ablations rather than local-validity strawmen: each checks a useful subset of the join and omits a different coordinate. Their false-allow counts are reported in Section 6.3, together with the analytic identity that governs the majority profile, because the aggregate counts are functions of corpus design.

5.6 Companion evidence-class upgrades

The main 104-transaction corpus retains corpus-supplied experimental authority-validity flags so that its revision lineage remains comparable. Eight companion laboratories change evidence class without silently reinterpreting that corpus.

Native signed authority/effect adapter. Eight deterministic fixtures use real Ed25519 verification for every delegation edge and the effect receipt. The verifier checks parent digest, principal and role lineage, permission, status, freshness, scope attenuation, terminal action scope, receipt issuer, chain digest, effect time, and action binding. One fixture passes; seven isolate a bad edge signature, bad receipt signature, signed action mismatch, expired edge, signed wrong lineage, signed scope amplification, and signed effect-time mismatch. The builder uses deterministic test-key seeds and persists public keys only. This is a Tyche experimental profile, not conformance with WAVE, JEDI, OAuth, DID, or another external credential format.

Mutation-generated transactions. Starting from one independently passing transaction, a deterministic builder applies 37 single faults, one at a time, across policy rules P1–P5, evidence rules E1–E2, state rules S1–S3, authority rules A0–A19, and measurement rules M1–M3. The transaction packet, both evaluator sources, and both unsealed outputs were hashed before comparison with the author-written mutation oracle. A second packet composes 12 cases with two to five simultaneous faults, including two within-layer precedence cases. The freeze establishes ordering, not label truth: generator, oracle, and labels remain programme-internal.

Prospective signed-revocation races. Thirteen scheduled event sequences place signed status snapshots before appraisal, between appraisal and commit, at the commit instant, and after commit, and add stale, lower-sequence replay, credential-substitution, status-unavailability, re-issuance, credential-window, and bad-signature cases. The strict rule verifies signed credential and status objects, uses the half-open credential window `valid_from <= t < valid_until`, requires fresh status at appraisal and commit, requires a non-decreasing signed sequence number, and treats revocation at the commit instant as effective. Independent Python and Node implementations evaluate the frozen packet. Three ablations omit commit rechecking, fail open on missing commit status, or ignore sequence monotonicity.

Native runtime-attestation appraisal. The lab embeds the public frozen Tyche `aep-pcr16-vector` at commit `2d22013987e1e98f8f9132832c9db2b967035945`, including its RSA attestation public key, `TPMS_ATTEST`, TPM signature, and PCR-values blob. The verifier recomputes

```
PCR16 = SHA256(0x00*32 || outcome_digest_bytes)
preimage = "tyche.aep.qual.v1|capsule=" || capsule_hex
           || "|outcome=sha256:" || outcome_hex
           || "|nonce=" || challenge_hex
qual_hex = HEX(SHA256(ASCII(preimage)))
checkquote_q = HEX(ASCII(qual_hex))
```

and invokes `tpm2_checkquote` with the last value, matching the frozen vector's ASCII-hex convention. One affirming case and six author-written cases, hash-pinned before the recorded invocation, cover outcome substitution, fresh-challenge replay, capsule substitution, declared-PCR substitution, quote-message tampering, and quote-signature tampering. The source vector was produced by `swtpm`; no hardware endorsement or live workload is inferred.

Live concurrent revocation service. A `ThreadingHTTPServer` listens on an ephemeral loopback TCP port. Under one lock it linearizes signed status reads, revocations, guarded commits, and unguarded effects, assigning each operation an increasing index. Every status-bearing response is Ed25519-signed and verified by the client. Four profiles are crossed with revoke-before-appraisal, revoke-between-appraisal-and-commit, forced cache-expiry, revoke-after-final-read, and revoke-after-commit placements; each profile also receives 64 same-barrier revocation/commit trials. `ttl_cache` uses a real 50 ms cache-age check and refreshes after a forced 75 ms delay;

this deterministic case separates it from `single_read`. `atomic_guard` checks status and commits the effect within one locked operation. `double_read`, `single_read`, and `ttl_cache` make an unguarded effect after a fresh, appraisal-time, or cached status, respectively. A false allow is defined by the service record itself: the client accepted and the credential status at the effect's linearization point was revoked.

Native state transaction overlay. The retained structural objects define the frozen source oracle, but no `status` field enters the native adapter. For each transaction, a subject-derived capsule identifier binds the source-corpus hash, transaction identifier, canonical action, and observed effect. PCR 16 binds a transaction-specific runtime measurement, while qualifying data binds the capsule, measurement, signed validity interval, and unique 32-byte challenge. Eight freshly initialized `swtpm` roots issue 104 quotes: four RSA and four ECC AKs, thirteen transactions per root. After `tpm2_checkquote`, PCR, challenge, and subject checks, a frozen appraisal policy applies contraindication, signed-window, and reference-measurement rules in the same precedence as the source state vocabulary. The verifier then re-runs the shared composition rule with only the state result replaced. One representative vector per root receives eight predeclared mutations: quote message, signature, PCR blob, challenge, transaction identifier, source corpus hash, measurement, and signed window.

An independent compiled verifier reprocesses the same frozen native packet in OpenJDK 21. It imports and executes none of the Python adapter, structural corpus, corpus builder, expected-label module, or saved mutation generator. It implements its own strict JSON parser, canonical transcript, SHA-256 and PCR derivations, appraisal precedence, and mutation construction, then invokes `tpm2_checkquote` directly. `jdeps` reports only `java.base`. The comparison therefore tests implementation diversity against the frozen packet, while remaining author-operated, same-host, and dependent on the same packet and TPM2 command-line verifier.

Process-isolated durable revocation service. Status and effect run as separate OS processes with separate TCP listeners. They coordinate through a SQLite append-only event log; every status read, revocation, and effect receives a durable linearization index and a signed response. The guarded effect checks status and inserts an idempotency-keyed effect in one `BEGIN IMMEDIATE` transaction. Simultaneous revoker and committer contenders run in separate client processes. In addition to the five scheduled placements and 64 same-release trials per profile, the runner disconnects after commit request but before response consumption, retries, and in a second fault class kills and restarts the effect service before retry. The primary safety oracle is the status stored in the effect transaction; the effect table supplies the duplicate-effect oracle. The experiment remains same-host and uses SQLite as a centralized linearization mechanism.

Containerized durable-revocation replay. A pinned offline image runs a status service, two interchangeable effect-service instances, a deterministic response-fault proxy, and the experiment runner as five distinct read-only containers over an internal bridge. A named-volume SQLite WAL store remains the centralized linearization mechanism. The proxy injects request loss before forwarding and response loss after durable commit. Recovery retries through the second effect instance and replays both allowed and denied idempotency keys. A separate saved-result verifier rechecks every signature, reconstructs the false-allow predicate from stored effect-time state, checks unique decision/effect keys, contiguous event identifiers, database integrity, and distinct signed service hostnames. It imports the service's canonical JSON serializer for signature preimages, so serialization is shared; the safety, idempotency, ordering, database, and hostname assertions are reconstructed separately. Container separation changes process, filesystem, network-namespace, and service-instance boundaries, but not the physical host, kernel, CPU, datastore trust domain, or operator.

Cross-ecosystem typed transfer. A pinned, field-minimized snapshot of the public EATF decision-path result supplies 21 rows: two accepts and 19 rejects covering 16 native rejection codes, with exact TypeScript/Python agreement. Every native code and every native first gate from the 104 transactions must appear in an explicit total crosswalk to eight coarse analytic classes. The builder exits non-zero on an unmapped code or a source-oracle mismatch. Native codes remain the primary result; the crosswalk is an analysis layer. A separate SQLite implementation consumes the frozen JSON mapping instead of the Python dictionaries and performs a leave-one-out plus wrong-class mutation for every mapping entry. This tests execution-path agreement, totality, and sensitivity, not the semantic correctness of the ontology.

A separate blind semantic-validation kit converts the 46 native states into opaque mapping tasks, supplies nine class definitions (pass plus eight rejection classes), a `NO_FIT` response, and a schema, and excludes the author mapping and all result fields. The author mapping is hash-sealed outside the four-file annotator archive. A coordinator validator and analysis plan are frozen before dispatch. The r6 execution package additionally supplies a deterministic native-documentation bundle, two-response validation, unweighted-kappa analysis with a preregistered 0.60 stop rule, independent adjudication schemas, and fail-closed checks for duplicate tasks, low agreement, residual `NO_FIT`, and unresolved `taxonomy-defect`. Its positive and negative self-tests pass while reporting `author_mapping_opened=false`. This closes the executable-study-design gap for semantic review, but the kit remains undispatched and therefore supplies no independent semantic evidence.

Standalone finite composition model check. A dependency-free Java program reconstructs the Boolean indistinguishability, four-coordinate binding, and bounded revocation models without importing the Python verifier, corpus builder, expected labels, SQL crosswalk, or saved verdicts. It checks both possible Boolean-composer outputs against a matched/mismatched pair, enumerates all 16 agreement vectors, and exhausts the six one-read and four topologically valid two-read revocation schedules. Its acceptance assertions are structural properties of those finite models; they do not extend the proof boundary to arbitrary implementation code, consensus, hardware, or deployed workloads.

The human-labelling experiment is deliberately separate. Its 90-transaction predecessor packet was superseded by a label-isomorphic 104-case derivative. The builder replaces mnemonic transaction, nonce, policy, profile, and principal identifiers with opaque aliases; re-signs policy and evidence objects under a packet-specific deterministic Ed25519 key; preserves invalid signatures and replay membership; and re-evaluates every typed tuple. Its known-token audit reports no remaining mnemonic match and the transformation preserves 104/104 typed tuples. The response schema, sealed author labels, and adjudication protocol are prepared but the packet is undispatched. No external label, adjudicated result, or agreement statistic exists. The coordinator archive now contains a pre-registered analysis program that validates and hash-seals both responses, computes verdict kappa with a fixed 10,000-replicate paired bootstrap interval, preserves the uncollapsed gate confusion matrix, and enforces the 0.60 stop gate without opening author labels. Its synthetic self-test exercises both the continue and stop branches; it is pipeline evidence, not human-label evidence.

6 Results

6.1 Policy state changed the appraisal surface

The strict verifier matched 16/16 expected tuples and allowed only the correct-policy/good-evidence cell. The signed-policy presence profile matched 11/16 with five false allows: good evidence under stale, substituted, version-only, freshness-only, and digest-only policies. The missing-policy fail-open profile matched 10/16 and additionally falsely allowed good evidence with no policy, six false allows in total. Both baselines still validate signatures, so the tampered-policy vector is denied by all three profiles. The SQL re-composition matched the Python verdict and rejecting gate 16/16, with the previously dead `policy.signature` branch now live.

Profile	Expected matches	False allows
strict identity/version/digest	16/16	0
signed-policy presence only	11/16	5
missing-policy fail-open	10/16	6

All counts are counts on this designed corpus, and the 16/16 needs the same analytic/substantive split that Section 6.3 applies to the composed layer. The split was disclosed there and not here in the previous revision, which was inconsistent: the disclosure belongs at both levels or at neither. Each expected label in this laboratory has two parts. The typed policy and evidence results are hand-written from the declared vector condition — a vector declared `stale` is expected to yield policy `STALE` — and recovering that declared condition from the signed bytes is what the verifier has to do, so per-layer agreement is the substantive check. The expected `verdict` and the expected `first_rejecting_gate`, by contrast, are machine-generated by the corpus builder from a two-layer precedence ladder (policy before evidence, each typed result mapped to one gate label), and `run.py` re-implements that same ladder when it composes. Agreement on those two fields is therefore agreement between two transcriptions of one small function, and it can fail only if a transcription diverges. That is weaker than the composed laboratory, where a single shared module makes the corresponding agreement outright analytic, and it is stronger than nothing, because two transcriptions can in principle disagree. It is not evidence that the precedence order is correct.

6.2 Metamorphic and confidence results

All 15 SAFE checks passed: the nine implementation identities passed as required (they cannot fail absent a coding defect), and the six informative checks also passed. The integrated frozen values were:

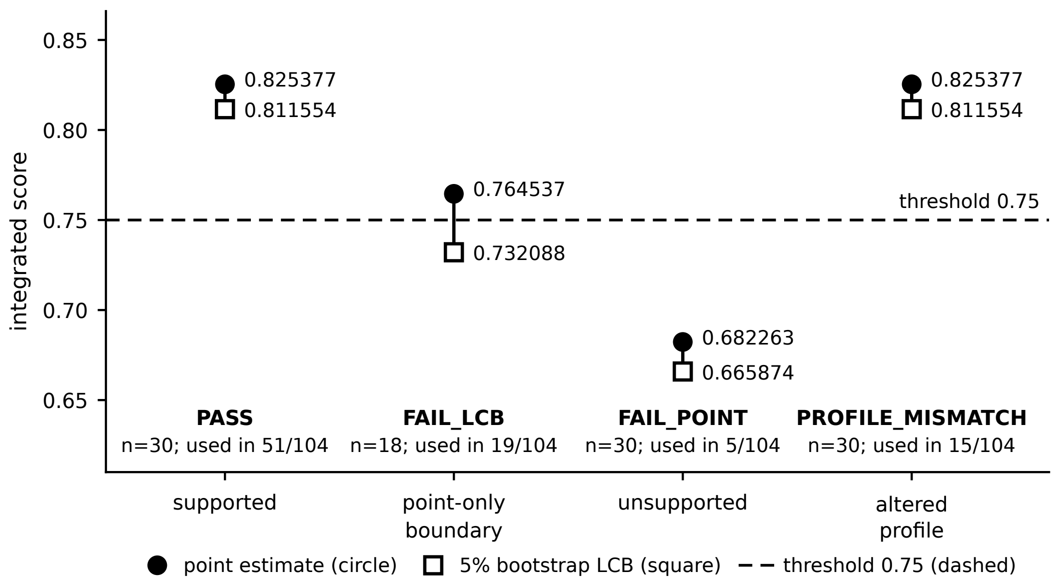
Operator	Tensor volume
geometric	0.7665940656
arithmetic	0.7691666667
RMS	0.7716991533

The harness accepts two-implementation agreement at a threshold of 10^{-12}; the observed gaps were smaller: exactly 0.0 for the three integration values and at most 1.1×10^{-16} (one unit in the last place) across the TOPSIS closeness scores. Six of six profile mutations changed the profile digest; these six mutations do not establish that the profile field set is complete.

The confidence fixtures produced:

Fixture	Point	5% bootstrap LCB	Threshold	Result
supported (n = 30)	0.825377	0.811554	0.75	PASS
point-only boundary (n = 18)	0.764537	0.732088	0.75	FAIL_LCB
unsupported (n = 30)	0.682263	0.665874	0.75	FAIL_POINT
altered profile (n = 30)	0.825377	0.811554	0.75	PROFILE_MISMATCH

A point-only gate falsely allowed two fixtures, and the two false allows are different phenomena: one confidence-gate false allow (the boundary fixture, whose point estimate clears the threshold while its lower bound does not) and one profile-gate false allow (the altered-profile fixture, which the point-only gate admits because it never checks the profile digest).



Designed boundary fixtures with author-chosen thresholds. The values are not measurements of any deployed model, and the n = 18 fixture sits where percentile-bootstrap bias is not negligible. The altered-profile fixture has scores identical to the supported fixture: its denial is driven by the profile digest alone. The four plotted fixtures cover 90 of the 104 transactions; the remaining 14 use two profile variants this laboratory owns in the same fixture schema, scored identically to the supported fixture and passed by the unmodified SAFE evaluator, which differ only in profile identity and are therefore not SAFE boundary cases.

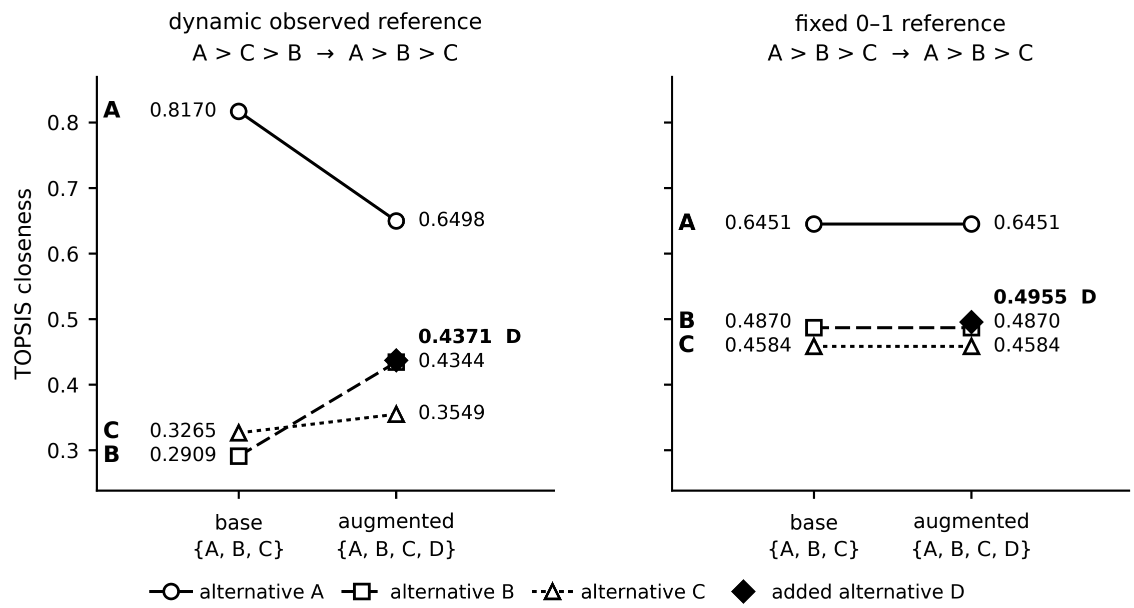
Fig. 3. The four confidence-aware measurement fixtures. Designed boundary cases with author-chosen thresholds; the values are not measurements of any deployed model, and the n = 18 fixture sits where percentile-bootstrap bias is not negligible.

In the overlap boundary fixture the threshold 0.737 was chosen by construction to sit between the two computed bounds: joint resampling of the shared columns produced LCB 0.732088 while incorrectly resampling them as independent produced 0.740781, so the decision changes at that threshold. This is a designed counterexample demonstrating existence, not a finding about any data-generating process; the dependence-sensitivity of bootstrap intervals is established statistics (Cameron et al. 2008; Künsch 1989; Politis and Romano 1994).

Dynamic-reference TOPSIS changed the relative order of the original three alternatives from A > C > B to A > B > C after the fourth alternative was added. Under the fixed 0–1 reference the base ordering was already A > B > C and was unchanged by the addition; the two reference schemes therefore disagree on the base ranking as well as on its stability, and the fixed scheme is not a control for the reversal but a different ranking function that is stable in this fixture. Rank reversal under reference-set change is a known property of TOPSIS and of normalisation-based multi-criteria methods generally (Aires and Ferreira 2018; García-Cascales and Lamata 2012; Wang and Luo 2009); the fixture reproduces the phenomenon inside this evidence profile, it does not discover it.

6.3 Integrated transaction results

The composed verifier reproduced all 104 expected typed tuples. That sentence requires its provenance statement before any interpretation: composition-level expected labels are generated by the corpus builder using the same shared composition-rule module the verifier imports, so agreement on the conjunction, the binding outcome,



A constructed reproduction of a known reference-set sensitivity of TOPSIS. The two schemes disagree on the base ranking, so neither is a control for the other: the fixed scheme is a different ranking function that happens to be stable here, not a scheme that retained the dynamic scheme's original order. Frozen v3 SAFE fixture, consumed read-only.

Fig. 4. TOPSIS closeness under the dynamic observed reference (left) and the fixed 0–1 reference (right), before and after adding alternative D. A constructed reproduction of a known reference-set sensitivity (Aires and Ferreira 2018; García-Cascales and Lamata 2012; Wang and Luo 2009); the two schemes disagree on the base ranking, so neither is a control for the other.

and the gate label is analytic, a property of one function applied twice. The non-analytic execution content of the 104/104 is the per-layer agreement: the re-executed policy, evidence, and measurement evaluators, the delegation-path evaluator, and the state adapter each reproducing the per-layer expectation carried by their source corpora. The subject-record agreement, also 104/104, is the stored `canonical_action` and `observed_effect` records agreeing field by field with the records re-derived from the artifacts at verification time, but both paths call `composition_rule.subjects_of`. It therefore checks serialization/plumbing and ensures the binding stage reads source fields rather than a stored verdict; it does not independently validate subject extraction.

Metric	Result
exact expected tuples (composition level, analytic)	104/104
per-layer expected agreement (substantive)	104/104
subject-record agreement, same-function round trip (plumbing)	104/104
allows	15
denies	89

Metric	Result
cross-layer denials: all five verifiers pass, composition denies	**8**
relational re-composition agreement	104/104
relational binding-outcome agreement (re-derived, not echoed)	104/104
source-corpus hash closure	PASS

The cross-layer-denial row is the article's strongest single number, and it is new: in every earlier corpus it was not merely zero but *structurally* zero, because the composition rule took six scalar strings and had no cross-layer argument, so no such transaction could be represented at all. The 8 transactions are:

```
XLB_effect_write_two_hop
XLB_effect_write_other_resource_two_hop
XLB_effect_export_four_hop
XLB_resource_other_one_hop
XLB_resource_other_two_hop
XLB_issued_before_window_two_hop
XLB_issued_after_window_two_hop
XLB_measurement_profile_two_hop
```

Each carries `failed_layers == []` and verdict `DENY`. The second, `XLB_effect_write_other_resource_two_hop`, is the counterexample this article opens with, verbatim: a WRITE effect on `invoice-999` under a path that permits only READ on `invoice-123`. Its sibling `XLB_effect_write_two_hop` isolates the effect alone, on the authorized resource, and `XLB_effect_export_four_hop` reproduces the same seam at four hops, so the result is not a property of path length.

The relational re-composition rows are a second code path, not independent validation: the SQL oracle consumes the per-layer typed results and the subject strings emitted by the Python evaluator and re-derives the conjunction, the binding outcome, and the first-rejecting-gate label. Because it re-derives the binding rules rather than echoing them, a divergence between the two implementations of those rules would be detectable; the subject strings it compares still come from the Python evaluator, so this remains a transcription check between two same-author implementations. It verifies no signature, no bootstrap replicate, no delegation edge, and no attestation object.

The ablations produced, in aggregate and per family:

Profile	Expected matches	False allows
strict typed composition	104/104	0
point-only measurement	102/104	2
four-of-five majority	73/104	31
artifact validity / policy presence	56/104	48
effect/resource partial binding	101/104	3
effect/resource + profile	102/104	2
effect/resource + time	103/104	1

Here `A` and `D` are strict allows and denies; `FA` is a false allow relative to strict composition.

8 of 104 transactions: every artefact verifier passes and the composition denies.

All 14 transactions of the cross_layer_binding family. In every row below the five artefact verifiers return policy PASS, evidence PASS, state PASS, authority ALLOW and measurement PASS, and the failed-layer set is empty; only agreement over the implemented effect/resource/report-time/profile coordinates varies. The binding stage is the join condition of the composition rule, not a sixth artefact verifier.

denied at the binding gate — 8 of 14	*binding result*	*verdict*
`effect_write_two_hop` reported ledger.write on invoice-123 · policy requires ledger.read · terminal grant [ledger.read] on [invoice-123]	**EFFECT_MISMATCH**	**DENY**
`effect_write_other_resource_two_hop` reported ledger.write on invoice-999 · policy requires ledger.read · terminal grant [ledger.read] on [invoice-123]	**EFFECT_MISMATCH**	**DENY**
`effect_export_four_hop` reported ledger.export on invoice-123 · policy requires ledger.read · terminal grant [ledger.read] on [invoice-123]	**EFFECT_MISMATCH**	**DENY**
`resource_other_two_hop` reported resource invoice-999 · authorised resource invoice-123 · terminal grant [invoice-123]	**RESOURCE_MISMATCH**	**DENY**
`resource_other_one_hop` reported resource invoice-999 · authorised resource invoice-123 · terminal grant [invoice-123]	**RESOURCE_MISMATCH**	**DENY**
`issued_after_window_two_hop` reported issued_at 2026-07-26T00:00:01Z · path valid [2026-07-25T00:00:00Z, 2026-07-26T00:00:00Z]	**TIME_MISMATCH**	**DENY**
`issued_before_window_two_hop` reported issued_at 2026-07-24T23:59:59Z · path valid [2026-07-25T00:00:00Z, 2026-07-26T00:00:00Z]	**TIME_MISMATCH**	**DENY**
`measurement_profile_two_hop` appraised under safe-profile-altered-v1 · policy requires safe-profile-main-v1	**PROFILE_MISMATCH**	**DENY**

matched controls — same construction, agreeing implemented coordinates **6 of 6 ALLOW**

`CONTROL_read_two_hop` · `CONTROL_read_one_hop` · `CONTROL_read_four_hop` · `CONTROL_window_lower_two_hop`
`CONTROL_window_upper_two_hop` · `CONTROL_main_profile_two_hop`

The binding stage is deterministic string agreement and interval containment over subject fields read out of corpus-supplied artefacts: it verifies no signature, takes no measurement, contacts no attestation service and is not attestation of any runtime. Its denials are denials on a designed corpus with author-written expected labels — coverage, not a rate, and not an estimate of how often such disagreement arises in any deployed system. Both window boundaries are inclusive, which is what the two boundary controls exercise. The controls are matched by construction and differ from the denials only in the reported subject fields, so a verifier that refused everything would fail them.

Fig. 5. The cross-layer denial class. The 8 transactions in which every artifact verifier passes – policy PASS, evidence PASS, state PASS, authority ALLOW, measurement PASS, and an empty failed-layer set – and the composition still denies, each shown with the subject pair that disagrees and the typed binding result that fired. Beneath them the 6 matched lookalike controls of the same family, built the same way with agreeing subjects, all of which are allowed: without them the denials would be consistent with a verifier that refuses everything. Counts on a designed corpus with author-written expected labels; the binding stage verifies no signature, takes no measurement, and is not attestation of any runtime.

Family	n	A	D	point FA	4/5 FA	validity FA
policy/evidence/measurement factorial	36	1	35	1	7	8
state–authority–measurement cube	8	1	7	1	3	7
authority path reuse	16	3	13	0	13	13
cross-layer joins	30	4	26	0	0	12
cross-layer binding	14	6	8	0	8	8
total	**104**	**15**	**89**	**2**	**31**	**48**

All three partial-binding errors occur in the 14-case cross-layer-binding family, where their omitted coordinates can be isolated:

Partial subject join	False-allowed mismatch class	False allows
effect/resource only	two time + one profile	3
effect/resource + profile	two time	2
effect/resource + time	one profile	1
full implemented effect/resource/time/profile binding	none	0

This ladder is the more demanding comparison: every row already requires all five local verifiers and some subject agreement. It shows the marginal coverage supplied by the implemented time and profile coordinates on the designed witnesses; it does not establish that these are the only coordinates a deployed transaction must bind.

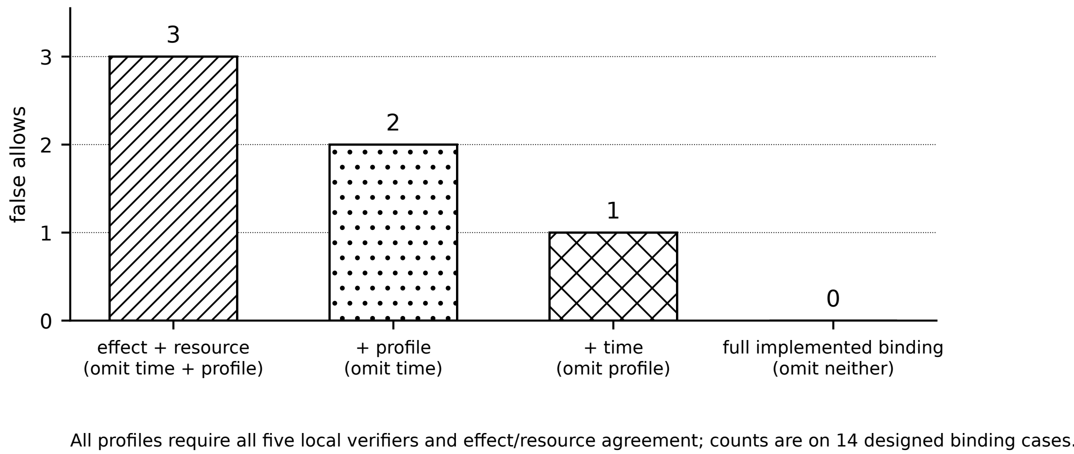


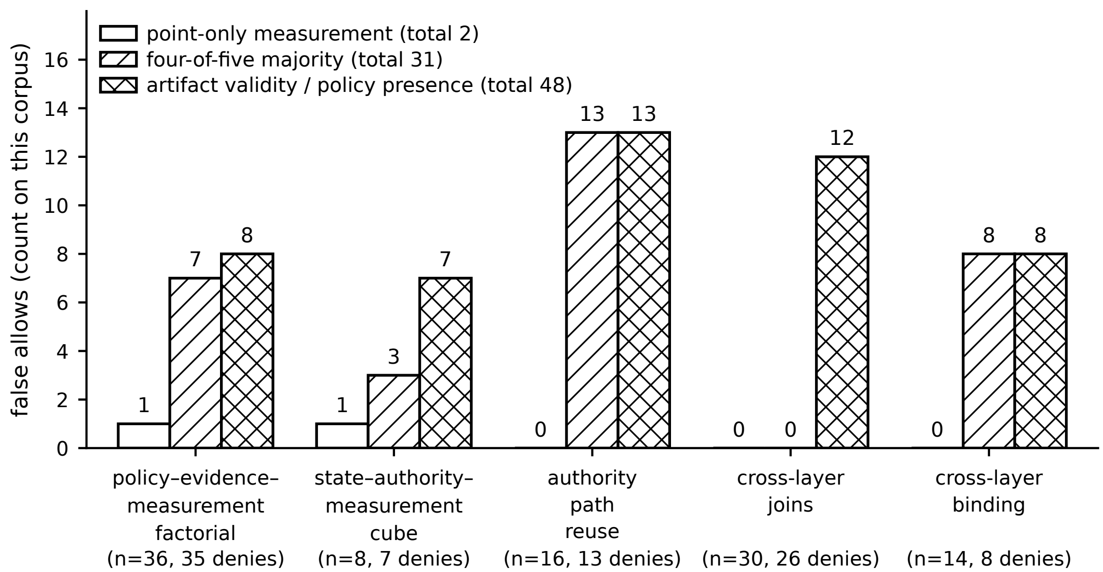
Fig. 6. False allows under progressively stronger partial subject joins. Every profile already requires all five local verifiers and effect/resource agreement. The remaining 3, 2, and 1 false allows isolate the coordinates omitted by each partial schema; the full implemented effect/resource/time/profile binding catches all 14 designed binding cases. Counts are corpus coverage, not rates, and the full schema remains explicitly partial relative to a deployed transaction.

Every verdict, gate, and co-failure count over the 90 transactions carried forward from the previous revision is unchanged, as are their artifact-validity (40) and four-of-five (23) false allows. One pre-existing number did move, and we state it rather than bury it: point-only false allows over those 90 fall from 3 to 2. The transaction is `F_correct_good_profile_mismatch`. The measurement evaluator's point-only result ignores the profile gate, so the previous revision's point-only ablation allowed a profile-substituted appraisal; the binding stage now catches it as `binding.measurement_profile`. This follows from the ablation definition in Section 5.5 — the point-only profile replaces the measurement criterion and keeps the binding stage — and it is a consequence of making the seam executable, not a re-tuning of the corpus.

The four-of-five column is still governed by an identity rather than a measurement, but it is no longer the identity the previous revision reported, and the failure of that identity is a result in itself:

	previous revision	this revision
four-of-five false allows	23	31
single-fault denies	23	23
cross-layer denials	0 (unrepresentable)	8
four-of-five = single-fault denies	holds	**false**
four-of-five = single-fault ∪ cross-layer	—	**holds**

A four-of-five majority allows a transaction if and only if at most one of the five artifact verifiers fails. Once denials with *zero* failing verifiers exist, its false allows are the single-fault denies **union** the cross-layer denials, 31 = 23 + 8, verified as exact set equality on transaction identifiers. Both statements are carried in the shipped summary, with the superseded one retained and explicitly marked as not holding rather than deleted. The single-fault layer distribution is unchanged: authority 14, policy 3, measurement 3, evidence 2, state 1. All 8 cross-layer denials are majority false allows because a vote over the five artifact verifiers cannot represent them at all — there is nothing for it to vote against.



Proof by construction plus corpus coverage on a designed 104-transaction corpus (89 strict denies). The four-of-five total (31) is the 23 single-fault denies plus the 8 cross-layer denials, verified as set equality on transaction identifiers, so it measures corpus composition and nothing else. A majority cannot represent a cross-layer denial at all: those transactions have zero failing verifiers. No bar is a rate or an estimate of behaviour in any deployed system. Ablations are author-defined experimental profiles, not representations of named external products.

Fig. 7. False allows by ablation, stratified over all five corpus families. Proof by construction plus corpus coverage: the four-of-five total is the single-fault denies plus the cross-layer denials by identity, no bar is a rate or an estimate of behaviour in any deployed system, and the cross-layer-binding bars show that a majority over five artifact verifiers cannot represent a denial in which none of them failed.

The artifact-validity profile inspects only policy presence and evidence integrity, so it necessarily false-allows every transaction whose faults lie entirely in state, authority, measurement, or the binding: that is 13 of 13 authority-reuse denies, 7 of 7 cube denies, and 8 of 8 cross-layer binding denies. The correct reading of the whole comparison is a proof by construction, that a compensating or incomplete rule cannot preserve a non-compensable specification, together with a corpus-coverage statistic; it is not a measurement of external systems.

Layer failures and co-failures are disclosed in full, and are reported separately from cross-layer binding because they are separate properties. Failed-layer occurrences across the 89 denies are policy 39, evidence 36, state 17, authority 30, measurement 39; state remains the least-exercised layer, and we say so rather than leaving the reader to notice. The predecessor corpus was block-diagonal, with zero co-failures in the four cells joining authority or state with policy or evidence. In the present corpus every off-diagonal cell of the co-failure matrix is at least 5, and 22 of the 31 possible non-empty failure subsets occur (previously 13). The cross-layer-binding family contributes nothing to any of these numbers, by construction: its 8 denials have no failed layer at all.

What that matrix does and does not establish should be stated plainly, because the previous revision's abstract and this section claimed that populating it exercised "the authority-with-policy and authority-with-evidence joins that the composition thesis names". It does not. A populated off-diagonal cell means two layers failed in the same transaction for independent reasons; it is co-occurrence, and it says nothing about whether the two layers

	policy	evidence	state	authority	measurement
policy	**39**	23	5	5	22
evidence	23	**36**	5	6	19
state	5	5	**17**	8	7
authority	5	6	8	**30**	6
measurement	22	19	7	6	**39**

Diagonal (bold) = transactions in which the layer failed; off-diagonal = transactions in which both layers failed, over 104 transactions and 89 denies. Every off-diagonal cell is at least 5: the corpus was engineered so that no cell is zero, so cell magnitudes reflect corpus construction, not failure prevalence anywhere. The matrix ranges over the five artefact verifiers only, so the 8 cross-layer denials — denials with an empty failed-layer set — appear in no cell of it. Counts on a designed corpus with author-written labels; they are not rates. State failures are structural-appraisal outcomes, not attestation verdicts of any runtime.

Fig. 8. Pairwise co-failure matrix over the 104 transactions. Counts on a designed corpus: the corpus was engineered so that no off-diagonal cell is zero, so cell magnitudes reflect corpus construction, not failure prevalence anywhere. Co-occurrence of two layer failures is not a join; the binding stage is not a layer and has no cell here.

were describing the same action. Co-failure coverage and cross-layer binding are two different properties with two different counts: the minimum off-diagonal co-failure cell is 5, and the number of cross-layer denials is 8. Neither number substitutes for the other, and only the second tests the seam.

The typed binding vocabulary is exercised as follows. Because the binding function is pure over the subject records, it is computed for every transaction, and corpus-wide it returns `PASS` 50, `EFFECT_MISMATCH` 28, `PROFILE_MISMATCH` 9, `RESOURCE_MISMATCH` 3, `TIME_MISMATCH` 3, and `UNDETERMINED` 11. Only a subset of those outcomes is *decisive*, that is, reached by the gate ladder because all five layers passed, and only those may be quoted as binding failures: `binding.effect` 3, `binding.resource` 2, `binding.time` 2, `binding.measurement_profile` 1, `binding.undetermined` 0. The larger corpus-wide counts are dominated by transactions whose policy is substituted or whose evidence is tampered — those deny earlier, and the binding outcome is never consulted —

and all 11 UNDETERMINED transactions carry a MISSING policy, which fires policy.missing first, so that branch is live in the vocabulary but never decisive here. The shipped summary keeps the two counts in separate fields so that the larger cannot be quoted as if it meant the smaller.

Coverage concentration is reported rather than hidden: one authority case (the exact two-hop positive path) backs 59 of the 104 transactions, and all 16 delegation cases appear. Six measurement fixtures appear: the four frozen SAFE fixtures, including the unsupported fixture, exercised five times, which makes the FAIL_POINT composition branch live, plus the two profile variants this laboratory owns for the binding family, which the unmodified SAFE evaluator passes and which differ from each other only in profile identity. The state layer has three failure outcomes and a pass, and all four occur: CONTRAINDICATED 9, STALE 4, and REFERENCE_MISMATCH 4, against 87 PASS. (The previous revision wrote "all four state failure outcomes"; there are three failure outcomes plus PASS.) Per-case usage tables are in the shipped summary.

6.4 Companion evidence-class results

The companion laboratories return the following exact designed-fixture results:

Evidence-class change	Executed result	Interpretation ceiling
native signed authority/effect adapter	8/8 verdict + first gate	real Ed25519 over a Tyche test profile; internal labels, no external-protocol conformance
single-fault mutations	38/38 oracle pairs; 570/570 Python/JS typed fields	1 baseline + 37 generated faults; frozen-before-comparison, but internally designed
multi-fault mutations	12/12 oracle pairs; 180/180 Python/JS typed fields	2–5 simultaneous faults; internally designed precedence cases
prospective signed revocation	13/13 oracle pairs; 13/13 exact Python/Node rows	scheduled sequences, not a live concurrent status service
native runtime-attestation gate	7/7 verdict-and-gate cases; four native + two binding negatives rejected	tpm2_checkquote over one frozen swtpm vector; no hardware or live workload
identified zeus2 vTPM companion	four-lane contract 20/20; 104/104 live TPM2 quotes; 64/64 native mutation rejects; no new transient or persistent handle	distinct Hyper-V VM/OS and Microsoft vTPM; no physical-host identity, manufacturer-certified EK, or hardware-root claim
live revocation service	276 traces, 890 persisted signed responses; atomic guard 0/0 safety errors	loopback HTTP and one process; safety result, not performance or multi-host evidence
native state transaction overlay	104/104 source classes and gates in Python and compiled Java; 64/64 mutation rejects and parity; 104/104 exact recompositions; Java 19/19 assertions; 8 distinct RSA/ECC AKs	cross-implementation native TPM2 evidence for the full corpus, but software TPM, shared packet, and one physical host

Evidence-class change	Executed result	Interpretation ceiling
process-isolated durable revocation	372 cases; 1,478 durable events; 96/96 fault recoveries; 0 duplicates; atomic guard 0/0	separate services/processes with restart and retry, but loopback and centralized SQLite on one host
containerized durable revocation	five containers; 135 decisions, 537 events, 95 effects; 540 signed responses; 4/4 fault recoveries; atomic false allows 0; duplicate decisions/effects 0; verifier 15/15	two effect instances and deterministic proxy, but one physical host and shared SQLite volume
typed EATF transfer	21/21 native-code rows plus 104/104 transaction rows mapped; SQL 125/125; 46/46 wrong-class and 46/46 omission detections; sealed blind 46-task semantic kit	analytic author-defined crosswalk; semantic kit undispatched; establishes no interoperability
standalone finite model check	both Boolean outputs; 16/16 binding states; 6/6 one-read and 4/4 two-read schedules; zero atomic-oracle mismatches	implementation-diverse exhaustive check of declared finite models only
isolated container re-execution	PASS for current 104-case corpus and the r3 evidence upgrades through scheduled revocation	different userland and interpreter; same host, kernel, CPU, and compiled NumPy/OpenSSL wheels
optional external-label extension	**not executed and not counted**	neutral 104-case packet and blind analysis pipeline prepared; no claim of human agreement is made

The revocation strict profile has no false allow against its 13 author-written expectations. The appraisal-only ablation false-allows seven cases: revocation between appraisal and commit, revocation at commit, stale commit status, lower-sequence replay, unavailable commit status, credential expiry at commit, and a corrupted commit-status signature. Commit fail-open false-allows the unavailable-status case; timestamp-only false-allows the lower-sequence replay. These values are intentionally case identifiers and coverage counts, not estimates of how frequently a deployed revocation service races.

The native runtime-attestation gate allows the unmodified vector and matches all seven hash-pinned author-written verdict-and-gate cases. Two negatives — substituted outcome and altered declared PCR — stop at the explicit measurement-binding precheck before the native verifier is called. Fresh-challenge replay, capsule substitution, quote-message tampering, and quote-signature tampering produce non-zero `tpm2_checkquote` returns. This closes the r3 review request for a native verifier call in a bounded companion experiment; it does not upgrade the structural state objects inside the 104 transactions by itself.

The native state overlay supplies that missing corpus-wide upgrade. All 104 quotes pass their cryptographic, PCR, qualifying-data, challenge, and subject checks before policy appraisal. The native results exactly reproduce 87 `PASS`, 9 `CONTRAINDICATED`, 4 `STALE`, and 4 `REFERENCE_MISMATCH` classes, 104/104. All eight AK public-key fingerprints, all 104 quote messages, and all 104 challenges are distinct. The 64 altered vectors fail at the native evidence gate, and substituting the native state classes into the original four other layer results yields 104/104 exact composed tuples. This closes the structural-main-state and one-vector limitations **within the designed**

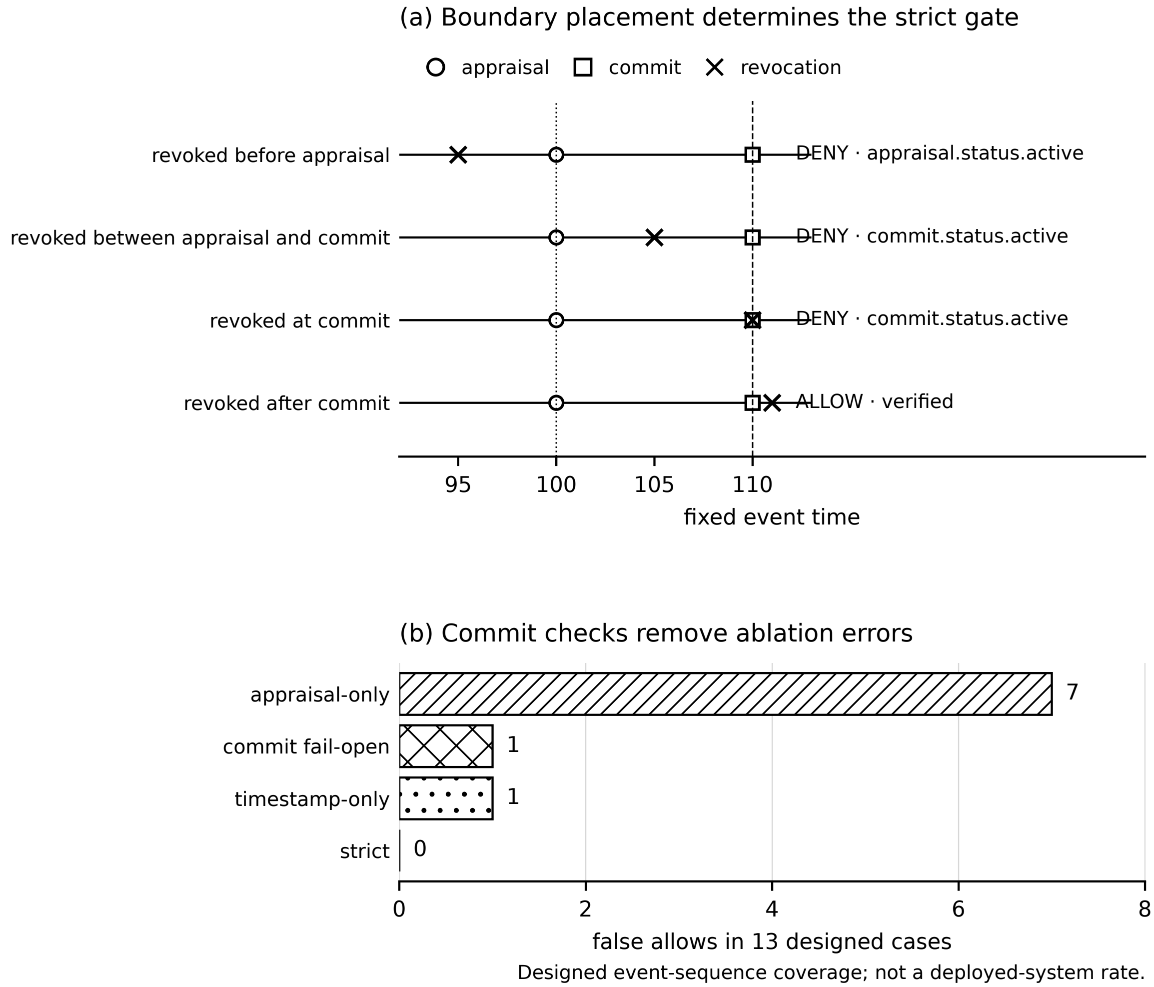


Fig. 9. Four temporal boundary placements in the signed revocation fixtures and false allows of three incomplete profiles. Revocation at the commit instant is effective under the strict profile; revocation after commit is not retroactive. Counts are over 13 designed scheduled sequences, not a rate or a live concurrency result.

benchmark. It does not attest a deployed workload: all roots are fresh software TPMs on the same x86_64 host, the reference policy is author-written, and no manufacturer endorsement or trusted external clock exists.

The independent compiled verifier matches the baseline state class and first gate on 104/104 vectors, rejects all 64/64 independently reconstructed mutations with exact state/gate parity, verifies 104/104 root-to-AK pins, and passes 19/19 predeclared assertions. The implementation source imports only `java.base`; its frozen results are bound to the exact overlay and primary verdict hashes. This removes a single-language/single-parser ceiling, not the shared-fixture, same-operator, software-root, or same-host ceilings.

The live service executes 69 cases per profile, including 64 same-barrier trials. Every one of the 276 result traces verifies its service responses; all 890 exact payload/signature pairs and the raw public key are persisted, and the saved-result checker reverifies them. The atomic guard has 0 false allows and 0 false denies. In this run the double-read, single-read, and TTL-cache profiles have 31, 39, and 31 false allows and no false denies. The forced-expiry case refreshes and denies under `ttl_cache` but false-allows under `single_read`, proving that the implementations are behaviorally distinct. The exact three incomplete-profile counts depend on operating-system scheduling in the same-barrier trials and are descriptive observations, not fixed expectations. Their deterministic revoke-after-final-read cases and Proposition 3 establish the existence of the TOCTOU schedule independently of those counts.

The process-isolated extension executes 93 cases per profile, 372 total, through separate status and effect service PIDs and separate contender processes. The durable log contains 1,478 events and 303 effects. All 372 case traces verify their signed responses. Across 96 response-loss/retry and kill/restart cases, 96 recover the original idempotent effect and none creates a duplicate. The atomic guard again has 0 false allows and 0 false denies; double-read, single-read, and TTL-cache expose 61, 66, and 63 false allows in this scheduling realization, with no false denies. Those three totals are not rates. The stronger observation is categorical: the atomic invariant and zero-duplicate invariant survive process separation, durable recovery, and ambiguous client outcomes, while every incomplete profile still has a stored counterexample.

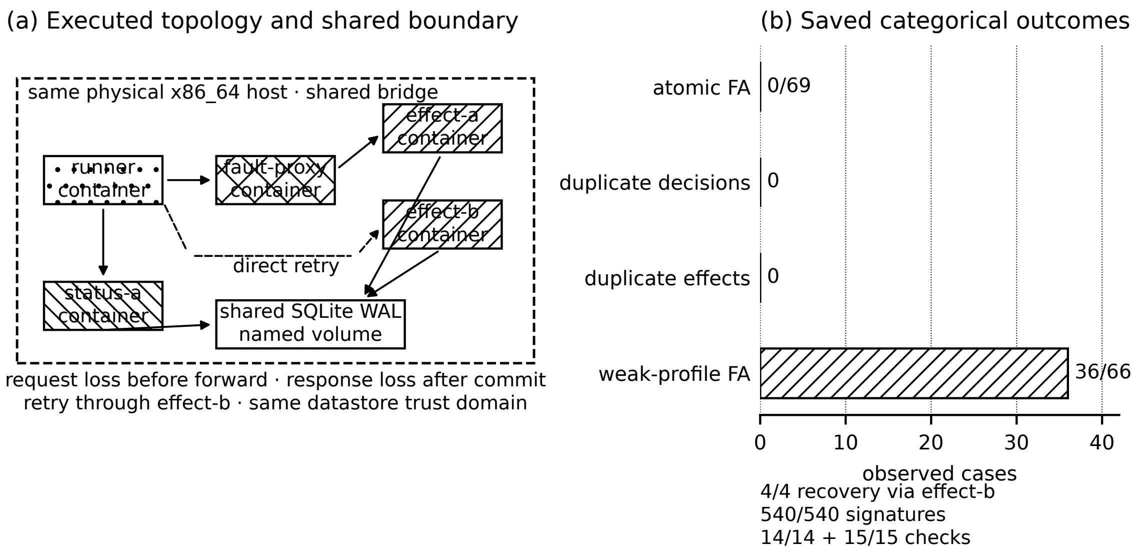
The five-container extension records 135 decisions, 537 events, and 95 effects across 69 atomic and 66 prior-read cases. All 540 persisted signed responses reverify. Four injected transport-fault cases recover through the second effect instance; allowed and denied retries replay the original idempotent decision, and no duplicate decision or effect key appears. The atomic profile has zero false allows; the intentionally incomplete prior-read profile exposes 36 in this scheduling realization, including its fixed scheduled witness. Fourteen in-run assertions and 15 separate saved-result checks pass. These counts add cross-instance and network-fault evidence on one host; they are not distributed-system or rate estimates.

The standalone Java checker accepts all of its finite assertions. For the shared all-pass local vector, each of the two possible Boolean-composer outputs makes at least one error across the matched/mismatched pair. The binding enumeration visits 16/16 agreement states: a Boolean-only rule admits all 15 mismatches; effect/resource binding admits 3; adding profile or time admits 1 in each branch; full effect/resource/time/profile binding admits 0. The revocation enumeration visits all six one-read schedules and all four topologically valid two-read schedules. The atomic oracle has zero mismatches; the incomplete profiles retain the canonical `R<X<C` and `R1<R2<X<C` false-allow witnesses.

The typed-transfer builder maps every native state and refuses partial crosswalks. The EATF result preserves 16 rejection codes across 21/21 TypeScript/Python rows; the transaction result preserves 28 rejection gates across 104/104 rows. Five of eight coarse analytic classes appear in both corpora. If only the Boolean `REJECT` were retained, the most frequent native code would identify 4/19 EATF rejects (21.1%) and the most frequent gate would identify 14/89 transaction rejects (15.7%); more fundamentally, 498 unordered pairs of distinct native rejection-code types become indistinguishable: 120 within EATF and 378 within the transaction corpus. These are code-type pairs, not observed errors, lost bits, or proof of semantic equivalence. The separately implemented SQLite path reproduces the coarse class for all 125 rows. Changing each of the 46 mapping entries in turn causes at least one mismatch, and removing each entry causes at least one unmapped row. This rules out a vacuous or unused crosswalk entry and adds a separate execution path; it does not make the mapping externally adjudicated.

The blind semantic-validation archive contains exactly 46 distinct native states and four declared annotator files. It contains no author class, expected class, oracle result, or validity flag; the frozen author mapping is represented only by a coordinator-side SHA-256 seal. Two consecutive archive builds are byte-identical and the response validator's synthetic self-test passes. Because no expert has received or returned the packet, these facts establish separation and readiness only, not semantic agreement.

The current offline container run regenerated the 104-transaction corpus with SHA-256 `1ba6d40d...a43a8`, the native signed fixtures with `09c868dc...583c8`, the single-fault Python answers with `3ce82a8c...92e0`,



Executed instance separation; one physical host and shared SQLite remain explicit ceilings.

Fig. 10. The executed five-container durable-revocation topology and saved outcomes. Panel (a) places the runner, deterministic fault proxy, status service, and two effect instances inside one dashed physical-host boundary; arrows show the solid initial request through the proxy to effect-a, the dashed direct runner retry to effect-b, and the shared SQLite WAL trust domain. Panel (b) reports zero atomic false allows, zero duplicate decision and effect keys, 36 designed prior-read counterexamples, 4/4 recoveries through the second effect instance, 540/540 signature verifications, and 14/14 plus 15/15 assertion sets. The physical host and datastore remain deliberately visible ceilings.

the multi-fault Python answers with `8596cb1d...0541`, and the revocation corpus with `33625c8d...c563`. Each full digest matches its host reference. Inside the container only the loopback interface existed, the route table was empty, and an outbound literal-IP TCP connection failed with `ENETUNREACH`. This closes the current **cross-environment** rerun, not the second-physical-host gate.

One negative provenance result is retained. The earlier independent JavaScript evaluator was frozen against `LABELLING-SPEC.md` version 1.0. The labelling-packet audit then corrected genuine specification ambiguities and bumped that document to version 1.1. Its old freeze therefore fails a post-freeze integrity check against the current packet. The 90/90 output is useful as a determinacy and implementation-diversity observation about the old specification, but cannot be presented as a clean independent evaluation of version 1.1 and, in any case, is not an external-human label result. The new neutral 104-case packet supersedes it for future human labelling but has not been sent to anyone.

7 Discussion

7.1 Policy is evidence about the appraisal procedure

The policy replay shows why policy should not remain ambient configuration. The same evidence class was allowed or denied depending on whether the verifier checked exact policy identity, version, freshness, digest, and signature. A signed policy was not enough: stale, substituted, and gate-isolated variants remained authentic

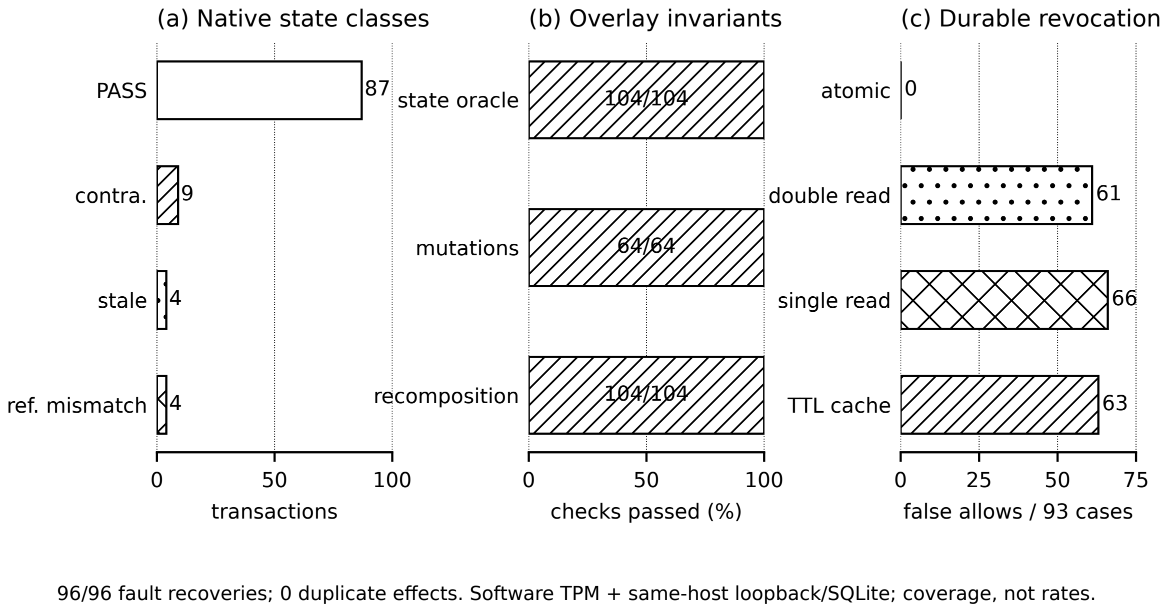


Fig. 11. The r5 evidence-class upgrades. Panel (a) reports the 104 native state classes. Panel (b) shows exact native-oracle agreement, mutation rejection, and recomposition. Panel (c) contrasts the atomic guard with incomplete revocation profiles over 93 cases each; all 96 injected fault cases recover and no duplicate effect occurs. Counts are designed same-host laboratory coverage, not rates.

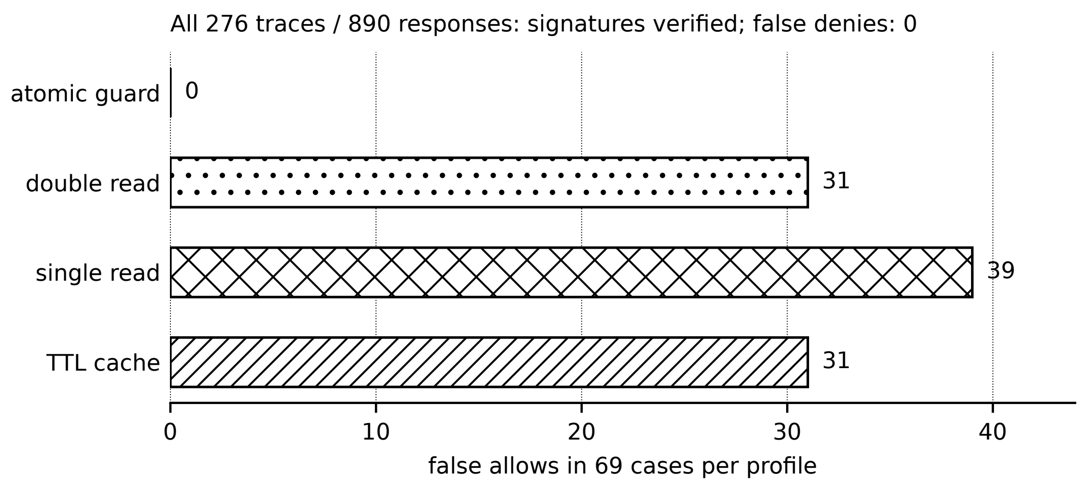


Fig. 12. Observed safety errors in 69 live concurrent cases per profile. The atomic status-and-effect guard has no error; profiles that separate status from effect expose false allows. Counts are one loopback run and not population rates or performance estimates.

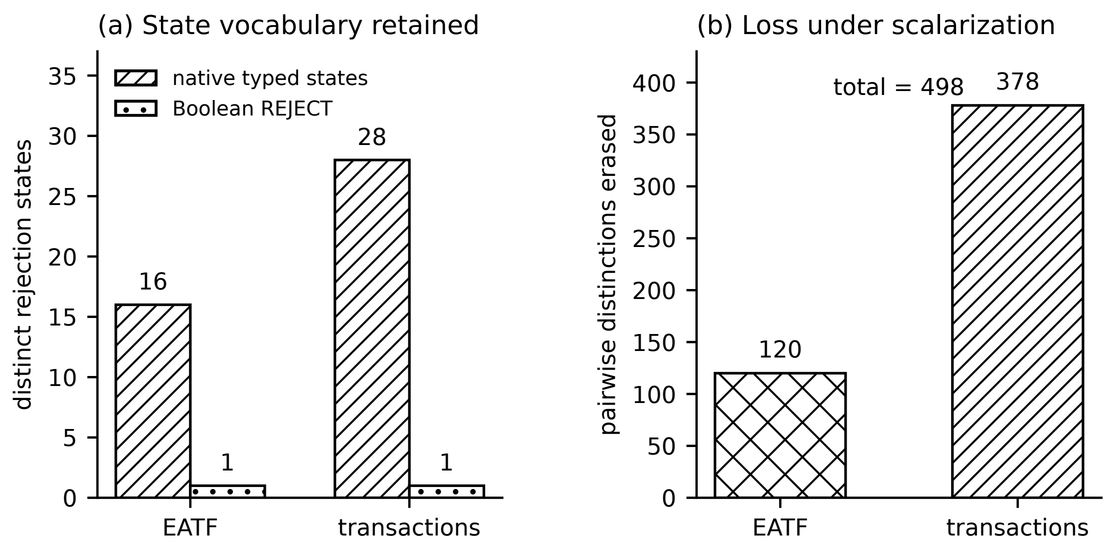


Fig. 13. Typed first-failure states retained in the EATF and transaction corpora, contrasted with one Boolean rejection state. The annotated pair counts are the native distinctions erased by Boolean scalarization, not error rates.

objects while being inapplicable to the intended decision, and the presence-oriented baselines admitted five and six of them.

Two policy fields changed status in this revision, and the change is what converts the section title from a slogan into a statement about executable behaviour. The signed policy object has always carried `required_effect` and `required_measurement_profile`. Until this revision they were inert: written by the policy corpus builder, read by no evaluator anywhere in the artifact, and load-bearing only in the sense that they were part of the bytes the digest covered. Changing `required_effect` would change the policy digest and therefore make the policy layer report `SUBSTITUTED`, but nothing in the system ever compared it to what the agent actually did. The claim that policy is evidence *about the appraisal procedure* was, at that point, supported by payload bytes rather than by behaviour.

Both fields are now read. `required_effect` is the source of `canonical_action.effect` and is compared against the effect the signed evidence reports and against the terminal granted tools; `required_measurement_profile` is the source of `canonical_action.measurement_profile` and is compared against the profile identifier of the fixture the measurement layer actually appraised. Each comparison has a named gate — `binding.effect` and `binding.measurement_profile` — and each is decisive on the corpus, 3 and 1 transactions respectively. The claim is now supported in the specific sense that removing either field, or writing a different value into it, changes a verdict rather than only a digest.

The honest ceiling is unchanged by this. What the binding stage compares are corpus-supplied strings; matching them establishes that two artifacts agree about a subject, not that either artifact is true. What changed is that a policy field now governs an outcome instead of decorating a hash.

The result supports including policy identity, digest, activation evidence, required effect, required measurement profile, and appraisal time in a verifier execution passport. It motivates those fields but does not establish their completeness.

7.2 Measurement support is not a scalar

Two fixtures had point estimates above threshold and still failed the strict measurement rule: one lacked a sufficient lower bound and one carried a different profile identity. The TOPSIS and overlap fixtures expose two more semantics that a scalar value omits: reference-set choice and sample dependence.

The contribution is not that every application must use one confidence level or one reference scheme. The contribution is that those choices must be bound to the evidence identity used to authorize an action.

7.3 Majority voting is the wrong composition

The comparison with the four-of-five majority is best stated as the identity of Section 6.3, in its corrected form: a majority admits every single-fault transaction by definition, and it also admits every transaction in which no artifact verifier failed at all, so on any corpus its false allows are the single-fault denies together with the cross-layer denials. No tuning of the corpus changes that logic. The second term is the more damaging one, and it is invisible to any composition that reduces the layers to a vote: a cross-layer denial has zero failing verifiers, so a majority over those verifiers has nothing to count. The design is intentionally adversarial to compensation because the predicates answer different questions. A valid policy cannot offset absent authority; an affirming attestation result cannot offset replayed evidence; a high metric cannot offset a substituted effect; and five passing verifiers describing different actions do not add up to one authorized action.

This is analogous to safety interlocks rather than ranking. A scalar may rank alternatives after every critical predicate passes, but it should not purchase permission to ignore a failed critical predicate.

7.4 Exact verifier identity remains a dependency

An earlier verifier-provenance differential, recorded in an internal companion report whose public deposit is pending, found 11/11 contract matches for the current TypeScript and Python entry points and 8/11 for an older unversioned demo entry point, with three false accepts; a tracked-commit rebuild reproduced the outcome matrix, so its lineage stands as outcome replication PASS; complete execution provenance INDETERMINATE. This article therefore treats exact source and corpus hashes as required artifact inputs and claims nothing further from that record.

7.5 From benchmark seed to evidence-carrying passport

The transaction object suggests a future evidence package:

```
measurement passport
  + verifier execution passport
  + exact policy snapshot
  + runtime attestation result
  + authority path
  + canonical action record
  + signed reported-effect record (`observed_effect` in the current schema)
  + typed binding outcome
  + typed composition verdict
```

Building on the SAFE component and integration families of Kolesnikov and Giudici (Giudici and Kolesnikov 2025, 2026; Kolesnikov 2025), the measurement part can be made explicit as

$$\mathrm{MP} = (m, \Sigma, D, \mathcal{P}, \mathcal{A}, \tau, \alpha, t_{\mathrm{eval}}, t_{\mathrm{exp}}, v, \sigma),$$

where m is the component vector, Σ its uncertainty/dependence description, D the data reference, $\mathcal{P}$ the measurement profile, $\mathcal{A}$ the aggregation/reference-set semantics, τ the threshold, α the confidence level, the two times delimit evaluation validity, v identifies the implementation, and σ binds the record. This tuple is our operational evidence proposal, not a tuple attributed to the cited authors. An evidence-carrying decision package then adds the exact policy snapshot, native verifier passports, runtime result, authority path, subject records, decision clock, and typed binding/composition outcomes.

The package should preserve each native verdict and the protected subject it describes. The composed verifier then checks semantic joins rather than relabeling native verifiers as incorrect, and the 8 cross-layer denials are the concrete demonstration of why the subject must travel with the verdict: in each of them every native verdict was correct about its own object, and the package as a whole was still not a basis for the action. In this artifact the two subject records `canonical_action` and `observed_effect` are the minimal form of that requirement.

7.6 Revocation must meet the effect at one boundary

The live race changes the design conclusion. Rechecking status immediately before commit is useful but is not sufficient when the effect occurs in a different operation. The double-read profile still has the schedule `read(active) -> revoke -> effect`. The necessary interface is not simply "fetch fresher status"; it is a guard whose revocation check and effect share one linearization point, or an equivalent transactional protocol that makes the same invariant externally auditable. This is a systems-design result about the boundary between authorization and effect, not a claim that every deployment can place both under one lock.

The decision clock also clarifies what must be carried in evidence. Client timestamps alone cannot adjudicate a race with clock skew. A signed, monotonic service sequence or append-only linearization receipt can. A future passport should therefore record the revocation object, its sequence, the guarded-commit receipt, and their common state-machine identity.

The process-isolated extension narrows an important alternative explanation: the first live result could have been dismissed as one in-process lock protecting one in-memory object. The new result keeps the invariant when status and effect have different PIDs and listeners, after the effect service loses its response or is killed and restarted, because the safety boundary is the durable transaction rather than either process. It still does not show a distributed protocol: SQLite is the shared serialization authority, all processes use one host clock and kernel, and no partitioned multi-host state was exercised.

7.7 Typed failure is a portable interface, not a universal taxonomy

The EATF transfer supports a modest but useful generalization: different verifiers can preserve their native failure vocabularies while exposing the same outer shape — verdict, first decisive state, protected subject, implementation identity, and evidence digest. The coarse ontology allows comparison, but its mapping is contestable and must remain visibly separate from native semantics.

This is preferable to a forced universal error code. A monitoring system can route `temporal_freshness` or `subject_binding` at a coarse level while a domain specialist still sees `TSA_IMPRINT_MISMATCH`, `evidence.replay`, or `binding.effect`. The 498 collapsed distinctions quantify why retaining only `valid=false` is inadequate for diagnosis, repair, or reviewer audit.

The SQLite mutation audit strengthens a different question from semantic validity. It establishes that every mapping entry is live and that two execution paths agree on the declared mapping. Whether `TSA_IMPRINT_MISMATCH` and `binding.effect` should both be called `subject_binding` remains an interpretive claim until independent domain experts or executable external adapters corroborate it.

7.8 Native state evidence can replace a typed fixture without changing composition

The overlay is an evidence-class substitution experiment. The original transaction subjects, other four layer results, binding records, and composition rule are held fixed; only the state result is re-derived from a native quote and frozen appraisal policy. Exact recomposition 104/104 shows that this substitution is observationally conservative at the decision interface while making quote, PCR, challenge, time-window, measurement, and transaction bindings falsifiable.

That result does not transform `swtpm` into hardware. Its contribution is the interface pattern: a typed state result can be backed by stronger evidence without changing the composition algebra, provided the evidence binds the same transaction subject and preserves the typed failure vocabulary. Hardware TPM, manufacturer AK provenance, remote verifier sessions, and a deployed workload remain separate experiments.

8 Threats to Validity

8.1 Construct validity

The strict composition reflects the study's definition of a subject-bound decision over the implemented projection. Other systems may use different evaluation orders or distinguish additional layers. The result does not imply that the five layers plus the binding stage are a complete ontology, and the binding rules themselves are a design choice: four comparisons in a fixed precedence, over subject fields this artifact happens to carry. A deployment with a different notion of "same action" would need different rules, and the time comparison here is containment in a window rather than equality to one timestamp. The main corpus's three decision-relevant constants are unrelated, so it does not instantiate the full decision clock. The live revocation companion instead uses server-side linearization indices for its critical order (Sections 4.2 and 5.6).

The policy corpus uses a test Ed25519 key and an experimental policy profile. The authority objects are not production WAVE, JEDI, OAuth, or MachineMandate credentials, and the authority layer's cryptographic flags are corpus-supplied **in the 104-transaction main corpus**. The companion native adapter and revocation laboratory verify real Ed25519 signatures but use Tyche experimental credential/status profiles and deterministic test keys; they do not retroactively make the main corpus cryptographic or establish conformance with an external protocol. The frozen source corpus still encodes its author-written state oracle in structural attestation-result objects. The native overlay replaces the *evaluation* of that predicate with 104 quote/PCR/challenge/reference-policy appraisals and then recomposes, but it does not make those author-written reference policies external ground truth or attest a deployed runtime. Its eight roots are software TPMs created on one physical host; no hardware root, manufacturer AK identity, protected trusted time, or remote workload exists. The measurement data are synthetic boundary fixtures. The binding stage compares corpus-supplied subject strings and contains an interval test; it verifies no signature, takes no measurement, contacts no attestation service, and does not upgrade any corpus-supplied flag into a cryptographic fact. A cross-layer denial on this corpus demonstrates that the seam is representable and localizable, not that any deployed system exhibits it.

8.2 Internal validity

Expected labels and evaluators originate from the same research programme, and at the composition level the labels are generated by the same shared composition-rule module the verifier imports; that agreement is analytic. Per-layer agreement is the non-analytic execution check; subject-record agreement is explicitly a same-function plumbing check. The same split applies, in a weaker form, to the policy laboratory: there the expected verdict and expected rejecting gate are machine-generated by a precedence ladder that the runner re-implements, so agreement on those two fields is agreement between two transcriptions of one function (Section 6.1).

Pure-Python/NumPy agreement and the relational re-composition paths reduce implementation-path risk but do not create external label independence; only the delegation-path oracle re-derives verdicts from raw case objects, and even it shares case expansion and expected labels with the Python entry point. Three of its four gate-isolation cases exercise a branch by flipping the corpus-supplied Boolean that branch reads, which is branch coverage in two implementations at once and not verification of the property the branch names (Section 5.3).

The mutation and revocation expectations are also author-written. A pre-oracle freeze prevents moving evaluator outputs after the mutation oracle is opened, but does not make that oracle independent or correct. The revocation implementations share the same frozen packet and internal expected labels. The independent-JavaScript freeze against the external-label packet no longer validates the current labelling specification because that specification was corrected from version 1.0 to 1.1 after the freeze; the drift is reported rather than repaired retrospectively.

Both live services derive their oracle from the state stored by the same linearization mechanism that records the effect. This avoids client-clock guessing but is not an independent implementation of the service semantics. The durable extension narrows the one-process explanation, not the centralized-oracle explanation: both services share one SQLite database. The exact false-allow totals in same-release trials vary with scheduler order; only the zero-error atomic invariant, zero-duplicate invariant, fault recoveries, and deterministic counterexample placements are acceptance conditions. The EATF crosswalk is author-written. Completeness, SQL agreement, and mutation sensitivity are machine-checked, but the choice of eight coarse classes is an interpretive analysis rather than ground truth.

The composed verifier imports source evaluator code. The corpus hash closure prevents silent source-corpus substitution within the run. A companion container run (Section 9) reproduced every then-current experimental quantity under a different Linux distribution, CPython patch level, C library, and SQLite engine, but that container shared the host kernel, the host CPU, and the host's compiled NumPy and OpenSSL binaries. That earlier container evidence-upgrade rerun includes the current 104-transaction binding corpus, native signed adapter, mutation corpora, and revocation corpus; its cross-environment claim remains a same-host claim (Section 9).

Two subsequent hosted-runner repetitions verify the sealed capsule, apply the same hash-recorded overlay, and execute four declared lanes on separate GitHub-hosted job VMs reporting x86_64 and ARM64, with exact architecture-specific dependency resolution. They satisfy the frozen semantic contract on both architectures and remove the cross-architecture gap for policy replay, composed-corpus evaluation, typed transfer, and process-isolated durable revocation. They do not execute the native-TPM overlay, one-process live service, Java/model-check, or five-container labs on ARM64; identify the underlying physical hosts; use a hardware TPM; or supply outside-operator independence. The native-TPM, live-service, neutral-packet, and typed-transfer additions in r4 were also rerun locally and hash-checked. The r5 source capsule now vendors the previously external SAFE and boundary-seal trees plus an offline amd64 wheelhouse. A hash-bound JSON result and full log record a successful clean temporary extraction on the same x86_64 host. The r6 workflow verifies that exact 210-member source capsule, applies the declared overlay, installs the same pinned dependencies natively for each architecture, and has now completed its four-lane contract twice on both reported architectures.

8.3 External validity

The 104 deterministic transactions do not sample deployed agent populations. False-allow counts compare profiles on this corpus, are partly determined by corpus composition (Section 6.3), and must not be interpreted as real-world rates. The count of 8 cross-layer denials is likewise a property of the corpus we designed, not a frequency: it says that at least 8 distinct subject-disagreement shapes are representable and were exercised, and it says nothing about how often such disagreements arise anywhere.

The recorded A2A evidence adapter in the authority lineage is hybrid; the experimental authority overlay remains synthetic. Separate local status and effect services, concurrent revocation, response loss, and effect-service restart were executed; a five-container extension adds two effect instances, an internal bridge, and deterministic request/response loss, but shares one physical host and SQLite volume. No public or remote service, payment, robot, institutional workflow, distributed clock, network partition, or multi-host revocation event was executed.

The neutral external-label packet covers all 104 cases, its blind analysis pipeline is executable, and it remains undispatched.

8.4 Statistical validity

Bootstrap and TOPSIS cases were constructed to expose boundary behavior. They show existence, not prevalence. The bootstrap uses the plain percentile method with n as small as 18, no BCa correction, and no seed or replicate sensitivity sweep; the boundary decision could move under a different estimator. Across NumPy 2.4.4, 2.4.6, 2.5.0, and 2.5.1 on one processor the lower bounds did not move at all (Section 9), but that probe holds the CPU and its instruction dispatch fixed, so cross-build reproducibility of that third decimal is not established in general. The overlap fixture threshold was declared in the frozen corpus as a designed counterexample. Larger studies must preregister realistic data-generating processes, thresholds, and consequences of false allow and false deny.

The same-barrier and same-release live-race totals are scheduling realizations, not binomial estimates of a stable event probability. No confidence interval is reported for them because the trials are not independent draws from a defined deployment distribution. Their purpose is to exercise both linearization orders and preserve exact event traces.

9 Reproducibility and Artifact Statement

The revised entry points are:

```
labs/policy-version-evidence-replay/run.sh
labs/composed-transaction-corpus/run.sh
labs/formal-composition-modelcheck/run.sh
labs/protocol-valid-unauthorized/verify_corpus.py
labs/protocol-valid-unauthorized/verify_delegation_paths.py
labs/protocol-valid-unauthorized/run_sql_oracle.py
labs/prospective-revocation-races/run.sh
labs/native-runtime-attestation/verify_runtime_attestation.py
labs/native-state-transaction-overlay/run.sh
labs/live-revocation-service/run_live_races.py
labs/distributed-revocation-service/run.sh
labs/containerized-durable-revocation/run.sh
labs/cross-ecosystem-typed-transfer/build_transfer.py
labs/cross-ecosystem-typed-transfer/verify_crosswalk_sql.py
external-label-packet-neutral-104/build_neutral_packet.py
external-label-packet-neutral-104/analyze_responses.py --self-test
(
  cd release-preparation-2026-07-28/crosswalk-semantic-validation
  python3 validate_semantic_response.py --self-test
)
repro/verify_portable_source_capsule.py
repro/run_clean_capsule_replay.py

# companion trees retained in zeus-followup-2026-07-27
native-signed-authority-adapters/
  build_fixtures.py
  verify_fixtures.py
```

```
  compare_results.py
thesis-v4-mutation-tests/
  evaluate_mutation_corpus.py
  evaluate_mutation_corpus.mjs
  compare_mutations.py
thesis-v4-multifault-tests/
  run_python.py
  run_js.mjs
  freeze_and_compare.py
```

The composed laboratory's entry point runs, in order, the state-adapter self-test, the corpus build, the cross-layer binding acceptance test, the composed verifier, and the relational oracle, then verifies both checksum manifests from their owning directories. The acceptance test is a gate rather than a report: it exits non-zero if the Section 1 counterexample is allowed, if any of its five component verifiers fails on that transaction, if the binding result or gate is not `EFFECT_MISMATCH` / `binding.effect`, or if the matched lookalike is not allowed at gate `verified`.

The frozen previous packet's SAFE laboratory is consumed read-only by the working tree and vendored under its original relative path in the portable source capsule. Each run writes raw typed results, a summary, environment metadata where relevant, and a SHA-256 manifest with relative paths. The original corpus entry points and native TPM appraisal require no network. Both live-race entry points deliberately use HTTP bound only to `127.0.0.1`; the durable extension uses two listeners and a shared local SQLite file. They perform no external request. That the original source contains no network call is a declared property verified by inspection; that no external network was reachable during the r3 container re-execution is a measured property, and the two statements are not interchangeable.

On 2026-07-27 each deterministic r3 entry point was executed twice end to end against the corpus reported here: every regenerated result file and the complete standard output were byte-identical between the two runs, and all checksum manifests verified from their owning directories with relative paths, including the pinned hashes of the frozen SAFE sources. The frozen-vector native and typed-transfer labs are deterministic and manifest-checked. The generated native overlay, both live-race results, and service process identifiers are intentionally not byte-stable because fresh AK material, TPM clock fields, operating-system scheduling, and process identity are evidence. Their acceptance assertions are typed-class and recomposition parity, mutation rejection, atomic safety, idempotent recovery, signature verification, and exposure of at least one false allow in every incomplete profile.

A companion container reproduction re-executed the entry points offline from a base image pinned by digest, with no network namespace, a read-only root filesystem, read-only host mounts, and a writable tmpfs scratch copy. A first run documents the previous 90-transaction corpus and the expected environment-metadata divergences. A second run executes the current 104-transaction corpus and the evidence upgrades. It reproduces the current composition counts (104 transactions, 15 allows, 89 denies, 8 cross-layer denials), signed adapter 8/8, single-fault mutations 38/38, multi-fault Python results against the frozen JavaScript typed fields for 12/12 cases, and revocation Python results against the frozen Node rows for 13/13 cases. The six regenerated artifact/result hashes printed by the container equal the host references. Inside the container the only interface was `lo`, the route table was empty, and a literal-IP outbound connection failed with `ENETUNREACH`.

The pinned container changes Debian/Ubuntu userland, CPython 3.12.3 to 3.12.13, glibc 2.39 to 2.41, and SQLite 3.45.1 to 3.46.1. It shares the host kernel, CPU, and the same compiled NumPy and OpenSSL wheels. A secondary probe under four NumPy releases found identical lower bounds for the four measurement fixtures; the frozen SAFE corpus generator produced eight one-unit-in-the-last-place source-data differences under NumPy 2.4.x, none of which reached a result file because the composed entry point consumes the packaged SAFE corpus. Both container runs are cross-environment evidence only; the separate hosted-runner repetitions below supply the cross-architecture execution evidence.

A deterministic 22.1 MB portable source capsule now vendors the V4 laboratories, the exact sibling SAFE and boundary-seal inputs their relative paths require, reproduction scripts, and an offline CPython 3.12 linux/amd64 wheelhouse. The archive has 210 members: its embedded manifest covers 209 payload entries and is itself the final member. Relative-path safety, payload digests, and the outer archive digest all verify. A clean temporary extraction on `tyche-factory-1` reran the deterministic core, finite Java check, SQL crosswalk audit, 372-case durable service, saved five-container evidence verifier, and native overlay successfully with exit code 0. The archive and runner hashes, environment, pass sentinel, timestamps, exit code, and full stdout/stderr are frozen in a JSON result and companion log. That original record remains an honest same-host result. A declared release overlay subsequently ran the verified sealed capsule twice on GitHub-hosted x86_64 and ARM64 VMs. The overlay repairs execute bits on three archived run scripts and replaces one hard-coded `x86_64` claim-boundary label with `platform.machine()`; it changes no fixture, evaluator, event-processing rule, expected count, or pass/fail assertion. In the preferred repetition, both jobs verified 210/210 capsule members, satisfied 20/20 architecture-neutral semantic assertions, and agreed on all 11/11 cross-architecture comparison assertions. The exact result counts were identical: 16 policy vectors, 104 composed transactions, eight binding denials, 21 EATF rows, 104 transfer rows, 372 revocation cases, and 96 fault cases. A prior repetition also passed the same contracts. The runs install the same exact dependency versions into architecture-native environments instead of attempting to reuse the amd64 wheelhouse on ARM64.

The preserved hosted durable-result field that says `same physical x86_64/aarch64 host` is a legacy same-execution-domain topology label inherited from the runner; it is not physical-host attestation. The accompanying environment and claim-boundary records control the interpretation: one job VM per execution, with underlying physical identity unknown.

The preferred and repeat runs are bound by the following anchors:

```
preferred source commit:
  c85bc3aa5769a2e94386b2d43e40ef16fafe16ed
preferred complete downloaded-evidence manifest:
  8d2275f3fbbb60f98b3b36f7f514c1abdf48f313a4843d02e207cd510cf82c23
preferred comparison JSON:
  9e2ac18591a53c8a46fe73151aa7b1e681473b6eb32199380e98073225218b08
repeat source commit:
  428ce24b6d77b4b9691d5e42a94f0f39eacea918
repeat complete downloaded-evidence manifest:
  7b07c138f25083a46d46687522ad41afe91a596d133a06ac1782a8999a6dc62f
```

These records establish repeated hosted-VM and cross-architecture execution; they do not establish physical-host identity or outside-operator independence.

The configured `zeus2` SSH target supplies a complementary execution domain. Hashed machine and boot identifiers distinguish an Ubuntu 24.04 x86_64 Hyper-V VM/OS instance; DMI reports Microsoft Corporation / Virtual Machine / Hyper-V UEFI Release v4.1, and TPM fixed properties report manufacturer `MSFT`. The target verifies the deterministic public source archive, then satisfies the same four-lane contract: 16 policy vectors, 104 composed transactions, eight binding denials, 21 EATF rows, 104 transfer rows, 372 revocation cases, 96 fault cases, and 20/20 semantic assertions.

The non-destructive TPM companion first passes 11/11 static/input checks and 17/17 negative safety checks. The Microsoft vTPM rejects `tpm2_createak` below a generic transient endorsement primary with `TPM_RC_POLICY`; the failure occurs before any quote and is not hidden. A declared compatibility overlay instead creates an

	Policy Replay	Composed Corpus	Typed Transfer	Durable Revocation	TPM quote verification	TPM mutation rejection
Local x86-64 software TPM 8 fresh swtpm roots; one physical host	16	104	104	372	832/832	512/512
Hosted x86-64 job VM four-lane contract 20/20; reported architecture; physical host unknown	16	104	104	372	—	—
Hosted ARM64 job VM four-lane contract 20/20; reported architecture; physical host unknown	16	104	104	372	—	—
zeus2 x86-64 Hyper-V VM four-lane contract 20/20; identified VM/OS; Microsoft vTPM	16	104	104	372	**104/104**	**64/64**

Counts are deterministic contract checks, not deployed-system rates. vTPM evidence is not hardware-rooted attestation.

Fig. 14. Execution-domain matrix for the four executable lanes and TPM companions. Cells report designed corpus sizes or native verification/rejection counts, not deployed-system rates. The zeus2 row is an identified VM/OS with a Microsoft vTPM; it is not evidence of the underlying physical host or a hardware root of trust.

owner-hierarchy transient primary and an ordinary transient RSA signing key. Its encrypted private blob remains inside the temporary private directory. The final run verifies 104/104 quotes with `tpm2_checkquote`, one unique random challenge and qualifying-data digest per transaction, and rejects all 64 predeclared message, signature, PCR, challenge, transaction, corpus, capsule, and representation mutations. A read-only handle census returns zero new transient and zero persistent handles; the public bundle contains no `.ctx`, `.priv`, or PEM private key.

```
zeus2 provenance:
  8da49f87f6c0c589873d618786bee086f03d8030f3f360301118da07f51f86ed
zeus2 evidence manifest:
  ce20410880c05f4ef0a5a65c8fed9006e63217d5e482e4d1a8395c9ff8ef7dcd
TPM run manifest:
  9d0722edfcbd33cbc717995c224e8f0c8a55f663bd8ab212127e1cf23b33a7bc
TPM preflight:
  90ed2149c973f950aa3c3ce333c187143763f24cb2aa7a98dc1aab57c7048207
```

A separate blind-safe public artifact repository records Anton Sokolov as author, ORCID 0000-0003-2452-7096, and Tyche Institute as affiliation: https://github.com/tyche-institute/subject-bound-evidence-composition-artifact. Original code and schemas carry Apache-2.0 terms; original prose, figures, synthetic corpora, fixtures, and generated research data carry CC-BY-4.0 terms, with upstream notices preserved. Its `v0.2.0-rc2` release is built only from an exact allowlist and excludes the sealed capsule, expected labels, participant coordination, and internal review records. Thus authorship, licence grant, repository execution, hosted cross-architecture evidence, an identified second VM/OS run, and live Microsoft-vTPM evidence now exist. Physical-host and hardware-rooted attestation remain outside the claim, not missing support for a claim the paper makes.

A deterministic public-release sanitizer now accepts only a clean exact Git commit and a sorted 348-path allowlist, preserves component licences and notices, excludes blind/coordinator and unresolved toolchain material, and passes 11/11 fail-closed secret, identifier, mnemonic, undeclared-path, and licence tests. Two builds over the release-candidate tree and one build from a clean clone of the exact source commit produced byte-identical 350-member archives. The extracted archive verified its 349-entry manifest and replayed its four included lanes with 20/20 semantic assertions. A direct public-tree PR-head matrix (GitHub Actions run 30349843070) independently returned 20/20 on the job VM reporting x86_64, 20/20 on the job VM reporting ARM64, and 8/8 in the cross-architecture comparison; the comparison records the exact checkout commit. These are release-engineering and hosted-VM results, not outside operator reproduction. The versioned GitHub release carries the deterministic archive and SHA-256 sidecar; the Git commit, tag, and archive digest are content anchors. A DOI is not asserted.

The composed corpus records:

```
policy corpus SHA-256 (16 vectors, unchanged in this revision):
  8d1925bfe0459dc9f42e554cf63d01adbe801b6c628ff61908bde5f9d9dd45da

delegation-path corpus SHA-256 (16 cases, unchanged in this revision):
  2b93fe67c3199e5e15f80254b3377411edeff4045aa8b8014519f2365baab45e

SAFE corpus SHA-256 (frozen, unchanged):
  e5a09eab9a7e48bb8060a73a26583c2ad0cd1fd0f13e15a86cb3e15d1b253689

state-fixture corpus SHA-256 (unchanged in this revision):
  86feb7501b432fefb2cc55abfd843c738693db4e91bc40a5334136f8ef7c0810

composed transaction corpus SHA-256 (104 transactions, this revision):
  1ba6d40d07e62a862b98ec52ab5c189eb491f19e4e1e69a411fba73bbb9a43a8

  superseded 90-transaction corpus of the previous revision:
  300fcf3e52f11c5c9083b725bb4027bd47a7e9bfe9e0d86c2ca43af81d9f0e7c

shared composition-rule module SHA-256 (conjunction, binding stage, ladder):
  ac430575a00e0090434202d41c430e9fb9818924fc435ea766c64759fe93e61d

native signed authority fixtures SHA-256 (8 cases):
  09c868dc0efd4cad69f4c0cc569ca35f07d800cf0fabd745665a503e846583c8

single-fault mutation packet SHA-256 (38 cases):
  af226bac2a7223cb607a2eea075fc93831185c225c25dfcb2c717c99df2f63ab

multi-fault mutation packet SHA-256 (12 cases):
  5233577ad02101f17911cc646a30d5a02d910fc6e8588db212bf6ed7123efc0a

prospective revocation-race corpus SHA-256 (13 cases):
  33625c8dfd7b36da53a49b9306a0034b5529447a2e35e66d3a2c21eae385c563

native software-TPM frozen vector / result SHA-256:
  8b31e0f5081ca8b2cef47ad735daffb8aa0c5bfa7fda9c9c8a9f9ca82478bc57
  70b73cd540434a192d64ed11e1c446a1857747326712fd87419f17e8777937b6
```

```
native 104-transaction state overlay / summary SHA-256:
  33d36515f7f73af046bb897eeb549cf8ea4ec35c871e41c556f1f63bde4f0406
  c564cd44734bcdb4b5039ea5d13262c8658978bad796c20137f140fb1e08cf6a

live revocation-service result SHA-256 (276 cases, this run):
  f61af5b5be7fb2ba300dcd6cfe4ea3fadeacc137fac74843402dc6e465fc1dfa

process-isolated durable revocation result SHA-256 (372 cases, this run):
  1d3cb71b005d468309cdfaef9bacb46b2d3f47effc1d6aaebb8603fb7d4b1403

containerized durable revocation result SHA-256 (135 cases, this run):
  ca3afc42dac51cfb7d8d81a6d5bf9779cd7c622d9294f65f0e9c65f7783deada

standalone finite model-check result SHA-256:
  b51f28323c1cf01374433f3cc307d653e8462c969df3398afe7651cbc9e6aaab

independent Java native verifier source / assertions SHA-256:
  a5a674e4243fddbe063135a845f920f14a017ba3a96b42a875c6e20e649c10f6
  88a9f136d24d5e548e742c3dc1a241ddd771ef4fb992ee9c3ac346b4a72dc31f

cross-ecosystem typed-transfer result SHA-256:
  7bb3896fdf324405917e2ff7d76d76dd22de2bf43240e7a17f4b63a8006e0c90

separate SQL crosswalk audit SHA-256:
  ea455ec857e7563781ee930004ca27e99b70fb20083e106ec81212a0211ca2ec

neutral 104-case labelling packet / sealed labels SHA-256:
  fe56e226687cf38f31c6c692efd75245ac2a9ea6f474135f082883668b3da5e9
  23ae1a4b15f65a2cf75b5d56c758d9f9ed494cfe11a5333fa26e192ca2ccd0ea

blind external-response analysis program SHA-256:
  5ca8bbbdf1077ce8517c43aa0413aba081d22992db5d0db006e5d468f7f18bb7

blind semantic-crosswalk annotator archive SHA-256 (undispatched):
  9638bcbdde19bca574c753f0a7c12cefa5a9a099b10987c5516ae81f0d5328e5

executed x64/ARM64 hosted-runner workflow SHA-256:
  2ceccf8d290d8005362811d56e07218eeeb43d536ad7d044e83626f62ac06af9

preferred hosted-run complete evidence manifest SHA-256:
  8d2275f3fbbb60f98b3b36f7f514c1abdf48f313a4843d02e207cd510cf82c23

preferred hosted-run cross-architecture comparison SHA-256:
  9e2ac18591a53c8a46fe73151aa7b1e681473b6eb32199380e98073225218b08

repeat hosted-run complete evidence manifest SHA-256:
```

```
  7b07c138f25083a46d46687522ad41afe91a596d133a06ac1782a8999a6dc62f

portable second-host source capsule SHA-256 (executed through declared
x64/ARM64 release overlay):
  8c8919f9826381da289b73dfd35721938ad5aafea5cd3687b23187589a2d0386

same-host clean-capsule replay JSON SHA-256:
  8b864fa0b314915272a74994f95f55381a2f2c924461fb6b0e3cdc04656701e4

same-host clean-capsule replay log SHA-256:
  6234c7fb577ba6a6f44028f25d9e45ed80da96ab7abb03e8e75871b82aba63ed

current evidence-upgrade container log SHA-256:
  7accc59e8f325d03a3ba9a59f02b5a15b6daf2aa7108ff83237e703065689da6
```

The shared composition-rule module is hashed and recorded because it is the single implementation of the conjunction, the binding stage, and the gate ladder that both the corpus builder and the verifier import; its hash is what makes the analytic character of composition-level agreement checkable rather than merely asserted.

The sanitized artifact repository is public. The source/coordinator repository and active double-blind materials remain private and undisclosed.

10 Conclusion

The artifact study operationalizes a narrow proposition: individually valid or plausible artifacts are insufficient when a required global relation is absent from local verdicts. The executable result instantiates that proposition only for effect, resource, report time, and measurement profile; it does not bind the state/workload subject or prove one complete action identity. In the previous revision that proposition was asserted and the artifact could not express it; the composition consumed five scalar layer results, so no transaction in which all five passed and the transaction was still inadmissible under C could be built. It can now, and 8 such transactions exist in the frozen corpus, each denied at a named binding gate with every component verifier passing, each paired with a matched all-valid lookalike that is allowed. The article's own opening counterexample — a correctly signed receipt reporting a WRITE effect where the valid delegation path permits only READ — is one of them, and it was allowed by the verifier this article previously shipped.

Across the frozen fixtures, exact policy appraisal avoided the five or six false allows created by weaker policy profiles on the 16-vector corpus. Metamorphic tests exposed profile, uncertainty, dependence, and reference-set semantics absent from a point score. On the 104-transaction composed corpus, point-only, majority, and artifact-validity ablations produced 2, 31, and 48 false allows respectively; the majority count is an identity, but with the single-fault denies *and* the cross-layer denials rather than with the single-fault denies alone, and the failure of the narrower identity is itself the result. The subject-aware partial joins leave 3, 2, and 1 false allows when they omit both time and profile, time only, and profile only, respectively. The counts are corpus coverage rather than rates, and the composed verifier's 104/104 decomposes into an analytic composition-level component, 104/104 per-layer execution agreement, and a 104/104 same-function subject-record plumbing check, all stated in Section 6.3.

The evidence boundary is now sharper. Native signed authority/effect credentials, mutation-generated single- and multi-fault transactions, scheduled revocation sequences, native offline appraisal of a frozen software-TPM quote, 104 transaction-bound quotes under eight fresh software roots with a separate compiled verifier, a standalone finite model check, process-isolated and five-container durable revocation races, and typed EATF transfer with a separate SQL sensitivity audit are all executable. The live result isolates the systems requirement the scheduled fixtures only suggested: revocation status and effect need one linearization boundary, and the recovery result shows why that boundary also needs idempotent effects. The transfer result isolates one diagnostic cost of Booleanization: 498 unordered within-corpus pairs of distinct rejection-code types collapse to the same Boolean value.

The experimental ceiling is explicit rather than a queue of unearned future claims. The verified capsule plus declared overlay completes its four-lane contract twice on GitHub-hosted job VMs reporting x86_64 and ARM64, and once on the identified zeus2 Hyper-V VM/OS. The zeus2 Microsoft-vTPM companion adds 104/104 live quote verifications and 64/64 mutation rejections with a clean post-run handle census. These observations support portability across the reported execution domains and a bounded vTPM evidence path. They do not identify an underlying physical host, attest a hardware root, measure multi-host consensus, or estimate deployed-system rates.

External human labels remain uncollected because the scoped study does not use them as empirical observations: the designed corpus is evaluated against an explicit author-defined reference specification, and the result claimed is cross-implementation agreement plus counterexample coverage. The undispatched neutral packet is an optional future construct-validity extension. Similarly, the crosswalk is claimed only as an executable author-defined typed adapter whose code preservation and mutation sensitivity are measured; no external semantic consensus or interoperability claim is made. These two extensions may broaden external validity, but their absence does not invalidate the reported synthetic-benchmark result.

The resulting r7 manuscript is therefore a submission candidate for the narrow claim actually tested: local Boolean validity cannot substitute for typed composition over the implemented effect, resource, report-time, and measurement-profile coordinates, and revocation status and effect require one commit boundary in the implemented model. It remains a benchmark and systems result, not a claim of production validity.

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11 Reproducibility Checklist for JAIR

All articles

(1) All claims investigated in this work are clearly stated. **Yes.**
(2) Clear explanations are given how the work reported substantiates the claims. **Yes.**
(3) Limitations or technical assumptions are stated clearly and explicitly. **Yes.**
(4) Conceptual outlines and/or pseudo-code descriptions of the AI methods introduced in this work are provided, and important implementation details are discussed. **Yes.**
(5) Motivation is provided for design choices, including algorithms, implementation choices, parameters, data sets, and experimental protocols beyond metrics. **Yes.**

Articles containing theoretical contributions

Does this paper make theoretical contributions? **Yes.**

(1) All assumptions and restrictions are stated clearly and formally. **Yes.**
(2) Complete proofs or arguments are included for all theoretical claims. **Yes.**
(3) Informal explanations of the theoretical results are provided. **Yes.**
(4) Appropriate citations are given for theoretical tools and prior results, including linearizability and artifact-centric MAS verification. **Yes.**
(5) Novel claims are stated formally as a theorem, propositions, or corollaries. **Yes.**
(6) Notation is motivated and limited to the local-verdict vector, projections, binding predicate, decision rule, and decision clock. **Yes.**

Articles reporting on computational experiments

Does this paper include computational experiments? **Yes.**

(1) Required source code is in the licensed, versioned public artifact repository and release. **Yes.**
(2) The source code has a license allowing free reproducibility use. **Yes—original code and schemas are Apache-2.0.**
(3) The source code has a license allowing free research use. **Yes—original code and schemas are Apache-2.0.**
(4) Raw, unaggregated synthetic fixtures and generated evidence are in the public artifact release. **Yes.**
(5) Unaggregated data have a license allowing free reproducibility use. **Yes—original synthetic corpora and generated data are CC-BY-4.0, subject to preserved upstream exceptions.**
(6) Unaggregated data have a license allowing general research use. **Yes—original synthetic corpora and generated data are CC-BY-4.0, subject to preserved upstream exceptions.**
(7) Random-number generation and seeds are described sufficiently for replication. **Yes.**
(8) Execution environment and infrastructure are described, including relevant software versions and explicit VM, architecture, software-TPM, and vTPM boundaries. **Yes.**
(9) Evaluation metrics and their motivation are explained. **Yes.**
(10) The number of runs used for each reported deterministic result is stated. **Yes.**
(11) Unsuccessful or unsatisfactory experiments are not silently ignored; the specification-freeze drift and initial failed container assertions are retained. **Yes.**
(12) Analysis extends beyond one-dimensional performance summaries to uncertainty, metamorphic relations, typed gates, and stratified coverage. **Yes.**
(13) Algorithm and experimental parameter settings are reported. **Yes.**
(14) Hyper-parameter optimization is not part of the study. **NA.**

(15) Statistical hypothesis tests for stochastic noise effects are not the evidential design used for these deterministic constructed fixtures. **NA.**

Articles using data sets

Does this work rely on one or more data sets, including benchmark-generator outputs? **Yes.**

(1) Newly introduced corpora are included in the versioned public artifact with a research-use licence. **Yes.**
(2) The new data have a license allowing reproducibility use. **Yes—original data are CC-BY-4.0.**
(3) The new data have a license allowing general research use. **Yes—original data are CC-BY-4.0.**
(4) Literature and public-source inputs have appropriate citations. **Yes.**
(5) Literature-derived public sources are publicly available. **Yes.**
(6) New designed corpora, their construction, provenance, label status, and relevant statistics are described. **Yes.**
(7) Preprocessing, augmentation, mutation, joining, and bootstrap procedures are described. **Yes.**

Explanations

This is the review-format submission candidate, not a claim of acceptance or publication. The licensed public artifact is a sanitized, exact-allowlist release: coordinator records, expected labels, and active blind-study material are excluded. The source capsule plus a declared overlay completed its four-lane contract twice on GitHub-hosted job VMs reporting x86_64 and ARM64. The identified zeus2 Hyper-V VM/OS independently completes that contract and adds a 104-quote/64-mutation Microsoft-vTPM companion. None of these records identifies an underlying physical host or proves a hardware root. External human labelling and semantic-panel studies are optional external-validity extensions; no human-agreement or interoperability claim depends on them.